Independent diffraction-stack reanalysis of the Cocos Plate lowermost mantle

Python implementation and public-archive reconstruction of Hutko et al. (2006)

13 October 2026

Abstract

We implement spherical single-scattering migration of transverse S–ScS waveforms and repeat it with historical and later public-archive observations. Event information comes from USGS; station responses and waveforms come from EarthScope, NCEDC, and SCEDC. The reconstruction produces independent numerical images but does not establish a repeat of the published 100 km discontinuity step. Differences in event and station membership, phase selection, and unavailable receiver splitting corrections limit comparison. PREM migration, two reference phases, phase censorship, source deconvolution, double-array stacks, and point/source-location tests are implemented. Three-dimensional tomography corrections and published full-wave synthetic panels remain incomplete. We distinguish calculations from interpretations and give a panel-level record of both successful computations and failed attempts.

1 Scope and principal results

The main publication is Hutko et al. (2006), Nature 441, 333–336. The four comparison sets are (1) the published reference, (2) independently processed historical observations, (3) an expanded event set in the same Cocos imaging region, and (4) expanded observations and a wider image domain. The published reference is visually distinguished from generated results; it contributes no amplitudes, model grid values, or digitized input data.

Dataset	Used events	Used traces	Historical traces	Covered area (km^2)
Published	15	270	270	not tabulated
historical	9	80	80	1968798
modern_same	34	531	80	2377736
expanded	36	913	80	4507582

Covered area is measured on the spherical shell at 2650 km depth, using at least 20 valid paths from three events. It is an illumination measure. Limited azimuthal coverage can still produce broad migration smiles even where this count is high. The expanded same-region subset includes every retained historical observation; its bounce-point bounds apply only to newly added observations; the wider subset retains all accepted paths. Both use the same magnitude, depth, distance, filtering, and automated quality criteria. The historical source clusters contain 4 southern, 53 central, and 23 northern traces under our stated latitude-bin assumption, compared with the published 54, 180, and 36. Southern illumination is especially depleted. Consequently, failure to recover the step in this reconstruction is not a falsification of the published interpretation with its more complete, differently processed data.

The publication reports an approximately 100 km vertical offset, a southeast reflector 160–190 km above the CMB, and a northwest reflector 250–290 km above the CMB. We compare those written scales with independently calculated stacks. We do not infer agreement from using a similar colour scale. The automated peaks below are maxima searched over 100–400 km height and may represent sidelobes or noise. They are not estimates of a slab boundary unless their waveform coherence, location, and uncertainty warrant that interpretation. The published display extends above a weighted stack value of 17. Our display uses a sum divided by the maximum hitcount, with colour saturation at 0.15. The written specification does not uniquely define the original weight normalization, so these amplitude scales are not directly comparable. We retain both mean amplitude and hitcounts to make alternative normalizations explicit.

Dataset	Reference	South peak km	North peak km	Offset km	Offset minus 100 km
historical	ScS	391	391	0	-100
historical	S	211	301	90	-10
modern_same	ScS	181	391	210	110
modern_same	S	221	301	80	-20
expanded	ScS	181	391	210	110
expanded	S	211	391	180	80

Only 0 of 12 selected maxima have a positive lower pointwise 95% event-bootstrap bound. Several maxima lie at 391 km, near the upper edge of the search interval. The historical S-reference offset of roughly 90 km is therefore not evidence that the published 100 km step has been reproduced: it is reference-dependent and lacks a positively resolved peak pair. The median measured minus PREM ScS–S residuals are historical -2.79 s, modern_same -2.07 s, expanded -1.93 s. The fitted residual-versus-bounce-latitude slopes are historical -0.297 s/degree, modern_same -0.008 s/degree, expanded -0.133 s/degree. These fits describe this accepted sample; they do not isolate lower-mantle velocity from phase-picking and path effects.

2 Observations, assumptions, and forward quantities

The observable is transverse displacement between direct S and core-reflected ScS. The inversion assigns signed waveform amplitude to candidate scattering positions; it does not recover an absolute shear-velocity perturbation or a temperature field. We implement isotropic, single-scattering diffraction stacking in a spherical reference Earth. Multiple scattering, elastic conversions, azimuthal anisotropy, and finite-frequency triplication amplitudes are outside this imaging approximation. The interpretation of a folded slab is kept separate from the calculated image.

The scientific specifications are the main text and supplementary legends of Hutko et al. (2006), with processing details from Lay et al. (2004) and Avants et al. (2006). The Nature legends take precedence where these descriptions differ. They specify magnitudes 6.0–7.2, source depths 218–610 km, epicentral distances 65–85 degrees, and a final low-pass corner of 0.3 Hz. They normalize migration traces to S; the earlier stacking study instead normalized to ScS.

We identify historical earthquake dates from Table 2 of Thomas et al. (2004). That table contains 15 entries but repeats event number 7; the text says 14. The entry dated 30 August 1999 is identified with the USGS event of 30 August 1998: its latitude and longitude match the printed values exactly and its depth differs by only 0.5 km. We treat the printed year as a probable typographical error and preserve this inference in the catalog. Its preferred magnitude is 5.7, a second historical exception to the stated selection bounds. The USGS preferred magnitude of the identified 1991 event is 7.3, outside the published range. Historical identities are preserved and such catalog exceptions are flagged; the new-event query retains the written magnitude and depth bounds. The historical 23 April 2000 entry is identified with the 17:01 UTC event, consistent with the location in the published table, rather than its larger 09:27 UTC neighbour. No hypocentre used in computation is copied from that table. Magnitudes are USGS preferred magnitudes of mixed types, retained in the event tables. This is not guaranteed to duplicate the original CMT magnitude scale.

All event records and moment tensors are fetched from USGS. All waveforms and StationXML responses are fetched from EarthScope, NCEDC, or SCEDC. Every request records its URL, retrieval date, SHA-256 hash, size, and status; failures are saved rather than silently filled. Natural Earth coastlines are used only for geographic context. No author waveform collection, simulation output, code, or plotting script is consulted. Published figure pages are used only for comparison, never as numerical arrays or digitized model inputs.

Polet and Kanamori (2002) do report illustrative regional corrections: an approximately east–west fast direction and 1.2 s delay at PAS, and a Bay-area two-layer example with an upper 135-degree, 0.7 s layer above an east–west, 1.4 s layer. These are nonunique regional models; they do not identify the complete station/path correction set used by Hutko et al. We do not silently assign these examples to every station. Their existence means that the missing correction set is a limitation of this reconstruction, not an assertion that the literature contains no numerical anisotropy constraints.

The historical station list is an explicit approximation to California broadband coverage, augmented by named western US stations. For later events the acquisition also queries BK, CI, IU, II, US, and TA stations between 30–49 degrees N and 126–105 degrees W. Historical sites are retained; additional stations are sampled at one site

per network per one-degree cell to limit redundant downloads. This thinning is a new acquisition assumption, not a published selection rule. The expanded analysis includes the retained historical observations. The same-region expanded subset retains every accepted historical trace and requires newly added traces to have calculated ScS bounce points within 0–15 degrees N and 93–87 degrees W. The wider analysis uses all retained paths and masks plots with fewer than 20 contributing traces or three events. These thresholds describe illumination, not guaranteed spatial resolution.

2.1 Instrument correction, polarization, and source removal

Each gap-free BH three-component record is detrended, tapered over 3% of its length, and deconvolved to displacement using the epoch-appropriate archive response. The response prefilter corners are 0.003, 0.005, 4, and 5 Hz; the upper corners are reduced if required by the input Nyquist frequency. The instrument water level is 60 dB. Sensor orientations from StationXML determine the rotation to north and east. A fourth-order two-pass 0.01 Hz high-pass filter and an eighth-order two-pass 2 Hz antialias filter precede sampling at 5 Hz. These fixed choices replace the incompletely documented trace-dependent filters in the papers.

For backazimuth b, measured clockwise from north at the station, the transverse component is

$$u_T = u_N \sin b - u_E \cos b. \tag{1}$$

An optional receiver correction rotates to a fast direction ϕ, delays the fast component by $\delta t/2$, and advances the slow component by $\delta t/2$, then rotates back. This preserves the average arrival time. The correction is implemented and tested, but the supplied numerical correction file is empty: the allowed written descriptions do not provide a complete usable station table, and published splitting maps were not digitized. Therefore the observational images in this report have *no applied receiver splitting correction.* This omission is a material difference from the publication.

USGS moment tensors are transformed from the spherical up–south–east basis to north–east–down. The predicted SH radiation coefficient is

$$R_{SH} = \frac{\boldsymbol{e}_T^{\mathsf{T}} \boldsymbol{M} \boldsymbol{n}}{M_0}, \qquad M_0 = \left(\tfrac{1}{2} \sum_{ij} M_{ij}^2 \right)^{1/2}, \tag{2}$$

where $\boldsymbol{n}$ is the takeoff direction and $\boldsymbol{e}_T$ the transverse unit vector. Where tensors are available, S and ScS must have matching signs and $|R_{SH}| \geq 0.6$. Unavailable tensors are flagged rather than invented; accepted unverified records are reported separately. A coefficient range reported for the original CMT solutions need not hold for the current USGS tensor, so this check can remove an originally used event.

Phase peaks are measured on the 0.3 Hz low-pass displacement. S is sought within 15 s of its PREM prediction. Its residual defines a common time shift for the ScS search; ScS is sought within 15 s of that shifted prediction and at least 6 s after S, with the S polarity. Picks on a window boundary are rejected. This automated rule replaces unpublished manual picks and can confuse interfering phases. We require a minimum peak-to-RMS amplitude ratio of 3 for both phases, using the interval 90–30 s before picked S for noise. Colocated sensors appearing under different network codes are represented once. All choices and rejections remain in the trace audit.

For each event, up to ten candidates with the highest signal-to-noise ratios are aligned on ScS and normalized by its signed amplitude. At least three candidates are required. Their mean over $[-8, 14]$ s is cropped at the nearest zero crossings around its principal peak and tapered with a 25% Tukey taper. This isolates an empirical source wavelet w and limits inclusion of precursor energy. It is an assumed reproducible replacement for the original manual choice of simple ScS pulses. A minimum correlation of 0.65 with this wavelet is required.

For displacement spectrum U and wavelet spectrum W, the source-deconvolved signal is

$$D(\omega) = \frac{U(\omega) W^*(\omega)}{\max\{|W(\omega)|^2, \epsilon \max_{\omega'} |W(\omega')|^2\}} \, G(\omega), \qquad \epsilon = 10^{-4}. \tag{3}$$

Here G is a fourth-order, two-pass Butterworth low-pass filter with corner 0.3 Hz. The FFT is zero padded beyond the combined signal and source lengths; the wavelet zero time is explicitly placed at index zero. The deconvolved S and ScS peaks are repicked within 5 s, and $d(t)$ is divided by its signed S peak. The event supplements show both pre-deconvolution displacement and deconvolved signals, each separately S-normalized, with at least 80 s before S and 65 s after ScS. Display clipping at twice the S amplitude is labelled; saved traces are not clipped. These diagnostics do not replace expert manual acceptance.

2.2 Spherical ray integration and migration

The Earth radius is $R = 6371$ km and PREM's CMB depth is 2891 km. We use the isotropic tabulation of PREM supplied with ObsPy. For ray parameter p in seconds per radian, shear speed $v(r)$, and a monotonically upgoing leg,

$$\Delta(p) = \int_{r_x}^{r_e} \frac{p\, v(r)}{r^2 \sqrt{1 - [pv(r)/r]^2}}\, dr, \tag{4}$$

$$T(p) = \int_{r_x}^{r_e} \frac{1}{v(r) \sqrt{1 - [pv(r)/r]^2}}\, dr. \tag{5}$$

The endpoint e is a source or receiver and x a candidate scatterer. Only $p < \min r/v(r)$ is admitted; legs that would turn below x are absent. We split radial integration at every material boundary, endpoint, and 15 km interval. Sixteen-point Gauss–Legendre quadrature uses $r - r_x = u^2$ to regularize near-grazing incidence. One thousand ray parameters, clustered near the upper limit, define monotone travel-time curves. Linear interpolation onto a 0.02-degree distance table avoids separate ray tracing for every voxel. Direct S, ScS, and multiple predictions used for picks come from ObsPy TauP; the scattering-leg integrator is independently implemented.

The original image domain is $[-6, 20]$ degrees latitude, $[-103, -77]$ degrees longitude, and 2310–2910 km depth, at 0.2-degree horizontal and 10 km vertical sampling. For the last 19 km below the CMB, basal mantle shear speed is extended solely to map post-ScS time. It is not interpreted as a shear wave in the fluid core. The expanded domain is $[-10, 25]$ degrees latitude and $[-110, -70]$ degrees longitude at the same spacing. Geographic latitudes are treated as spherical; ellipticity, topography, and station elevation travel-time corrections are omitted. Profile endpoints are specified in configuration, independently of the published plots. Horizontal distances are measured at the CMB radius, with no vertical exaggeration. This differs from measuring profile length at the Earth's surface.

For reference phase q (S or ScS), the sampled time is

$$t_{iq}(\boldsymbol{x}) = t_{iq}^{\text{pick}} + T(s_i, \boldsymbol{x}) + T(\boldsymbol{x}, r_i) - T_{iq}^{\text{PREM}}. \tag{6}$$

The measured reference establishes a datum that reduces origin-time and shared path errors. Direct S and predicted sS and SS are multiplied by symmetric raised-cosine notches $m(t) = \sin^2\{(\pi/2)\min(|t - t_0|/2\,\text{s}, 1)\}$. The ScS-censored variants apply the same 4 s full-width notch to ScS. The exact published taper shape is not specified. Our hitcount counts valid paths and samples, including samples with tapered amplitudes, rather than a count of nonzero amplitudes. Consequently its retained/censored versions can coincide.

For $N(\boldsymbol{x})$ valid traces, both the arithmetic mean $\bar{A} = \sum_i d_i[t_{iq}(\boldsymbol{x})]/N(\boldsymbol{x})$ and the displayed hitcount-weighted mean are saved:

$$A(\boldsymbol{x}) = \bar{A}(\boldsymbol{x})\, \frac{N(\boldsymbol{x})}{\max_{\boldsymbol{x}'} N(\boldsymbol{x}')}. \tag{7}$$

This explicit interpretation of "weighted according to hitcount" is an assumption. No geometrical-spreading or absolute scattering-amplitude inversion is performed. The display saturates at ± 0.15 in S-normalized amplitude units; unsaturated arrays remain available. No positive reflector or step is imposed on observational images.

The first-order tomography correction operator is

$$\delta T = -\int_{\text{ray}} \frac{\delta v}{v}\, dt. \tag{8}$$

Its differential contribution is the sum on the two scattering legs minus that on the reference ray. The segment operator is implemented and verified for a uniform perturbation. The UT, UCB, and CIT coefficients are not specified by the written descriptions. We do not manufacture these models, download the authors' numerical inputs, or substitute generic velocity anomalies under their names. These three Figure 3 panels remain unreproduced.

2.3 Additional tests and the boundary of the reconstruction

The double-array test solves $\Delta_s(p) + \Delta_r(p) = \Delta_{sr}$ at each candidate reflector depth and samples each trace at that reflection time. ScS bounce latitude 6 degrees N separates southern and northern bins. Confidence intervals resample entire events 300 times, preserving within-event correlation. They are pointwise intervals, not corrections for searching across depths. The source-location sensitivity test perturbs only sources south of 25 degrees S by ± 40 km in depth or by 40 km toward/away from the selected profile midpoint.

Synthetic point tests inject one-second Gaussian arrivals from two explicitly assumed isotropic scatterers at 40% and 70% along profile A, at depths 2611 and 2711 km (100 km vertical separation). These test the diffraction-stack operator and acquisition geometry, not full elastic propagation. Each pulse has unit positive peak and a Gaussian standard deviation of 1 s; attenuation, conversions, and additional reflections are absent. The same two scatterers are tested with each dataset's own accepted event/station geometry. A separate kinematic SH test uses a 2.6% shear-speed increase 264 km above the CMB and optional 2.6, 5, or 10% decreases 90 km above it. With equal density, its local plane-interface displacement coefficient is

$$\mathcal{R}_{SH} = \frac{\mu_1 q_1 - \mu_2 q_2}{\mu_1 q_1 + \mu_2 q_2}, \qquad q_j = \sqrt{v_j^{-2} - p_h^2}, \quad \mu_j = \rho v_j^2. \tag{9}$$

Complex q_j permits a postcritical phase shift. This illustrative ray calculation does not include the refracted Scd branch, caustic amplitudes, attenuation, or full-wave reflectivity. It therefore cannot reproduce the published synthetic waveform panels. The 1.5% step model at 170/270 km above the CMB is generated as an input-model illustration on PREM. No 2.5-D finite-difference solver or thermomechanical slab-flow calculation is claimed.

There are no time-evolution initial conditions in the observational migration: its input is the measured, processed time series, zero outside its sampled support. The ray reference has a fluid-core SH reflection boundary and a solid surface as encoded in PREM/TauP. The kinematic examples assume causal localized impulses and isolated interface interactions; no absorbing numerical boundary or dynamic initial stress field is applicable to them. A full-wave reconstruction would require these additional specifications and validation.

3 Validation and numerical efficiency

Across nine source-depth/distance cases, independently integrated ScS times differ from TauP by at most 0.00170 s. A homogeneous spherical-shell test has maximum travel-time error 0.00718 s and angular error 0.000774 degrees, including nearly grazing rays. The zero-distance integral differs from independent radial quadrature by -1.77e-06 s. A known negative impulse of amplitude -0.3 relative to a positive impulse is recovered with ratio -0.29998; both peak times agree within one 0.2 s sample. Zero input remains zero. A splitting correction followed by its inverse has relative error 8.15e-16 for the chosen integer-shift test. The tomography segment operator gives the expected -0.3 s perturbation for a 1% anomaly over 30 s reference travel time. These checks validate units, signs, elementary timing, and indexing. They do not validate automatic picks on interfering real phases or the omitted receiver corrections. The point-scatterer test uses the same kinematics for synthesis and migration, so it is an operator/illumination check rather than independent validation of the Born approximation. The initial near-grazing quadrature failed the homogeneous-shell test; the final variable transformation and 16-point rule resolve that failure. All validation assertions are rerun by the rebuild command.

Runtime improvements are confined to substantial costs: six concurrent archive requests, event-level preprocessing workers, vectorized radial quadrature, travel-time tables, parallel independent datasets/events, restartable event checkpoints, and reuse of response and waveform downloads. Plain functions and NumPy arrays keep the seismological calculations readable. No GPU is used. The separate `usage.pdf` reports measured local CPU time, wall time, and process-tree memory alongside explicit assumptions for unobserved remote usage. An earlier long foreground build was terminated by the execution environment (exit 143) during the wider migration. Event checkpoints and event-level parallelism were then added. The historical and same-region weighted amplitudes, means, trace counts, and event counts were bit-for-bit identical to the saved serial results before the final dataset-membership correction (`logs/parallel-consistency.json`). The final documented rebuild reruns numerical checks and figure generation before compiling the PDFs; its stage outcomes are retained in `logs/latest-run.json`. A final membership audit found that the same-region filter had excluded six historical traces. The filter was corrected to retain the entire historical subset, then that dataset and its figures were regenerated.

4 Figures and comparison with the published reference

The generated sections have no vertical exaggeration and retain signed amplitudes. White regions fail the stated coverage threshold. Sidelobes and broad negative regions should not be interpreted automatically as low-velocity material. The receiver-correction omission, altered source wavelets, and strict use of current USGS radiation coefficients remove some originally contributing paths and can change the balance between northern and southern illumination.

A failed or unstable peak is retained as a failed comparison, not replaced by a synthetic reflector. The four figure sets follow; published panels are reference images only and are not counted as reproduced results.

Set 1: published reference only

The definitive original panels and captions are at https://www.nature.com/articles/nature04757. The reference plate below is cropped from the published PDF solely for visual comparison. It is not generated by this implementation. Copyright Nature Publishing Group (2006).

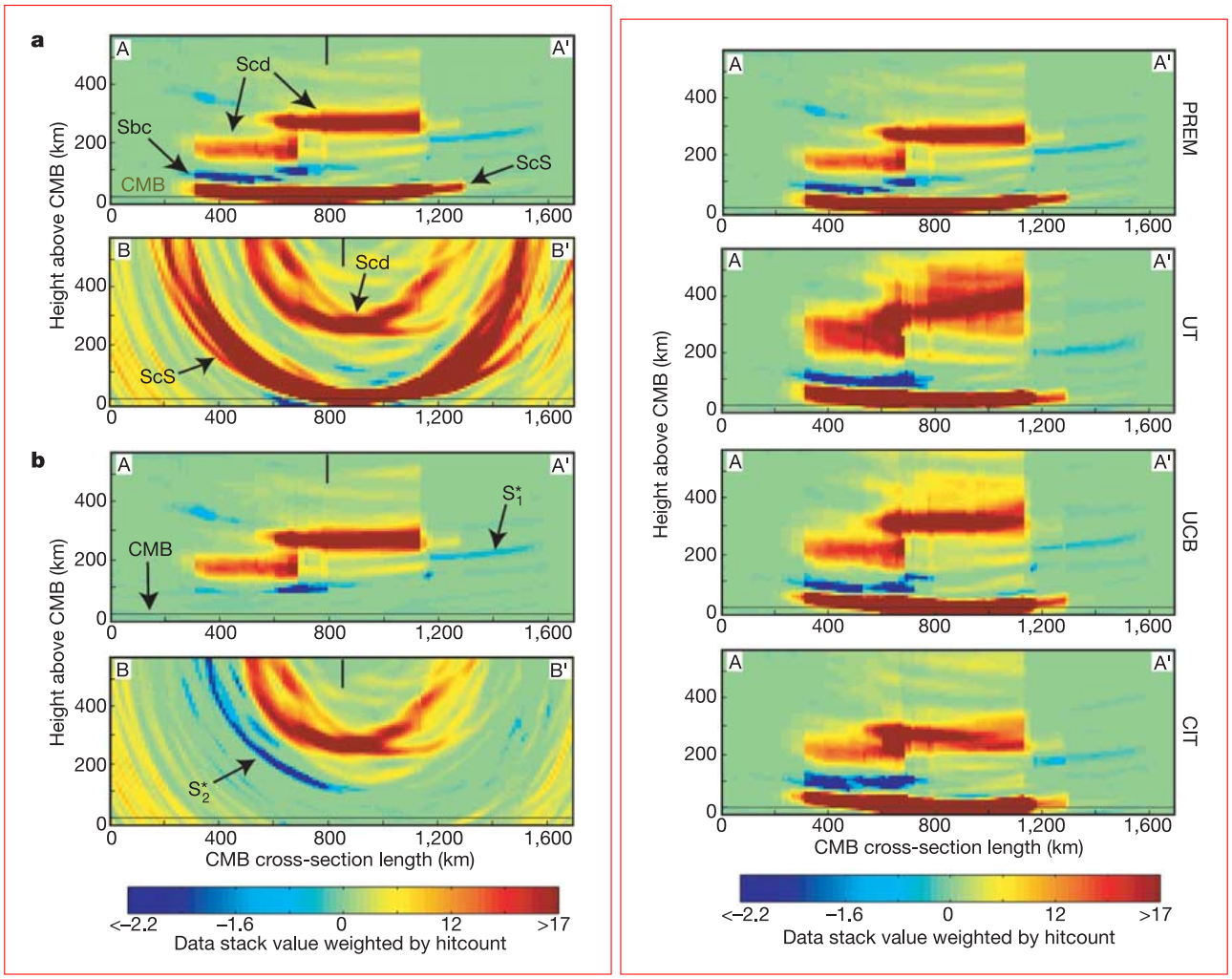


Figure 1: PUBLISHED REFERENCE ONLY: manuscript Figures 2 and 3. Red borders distinguish published images from every computed image in this report. Neither image supplies numerical input.

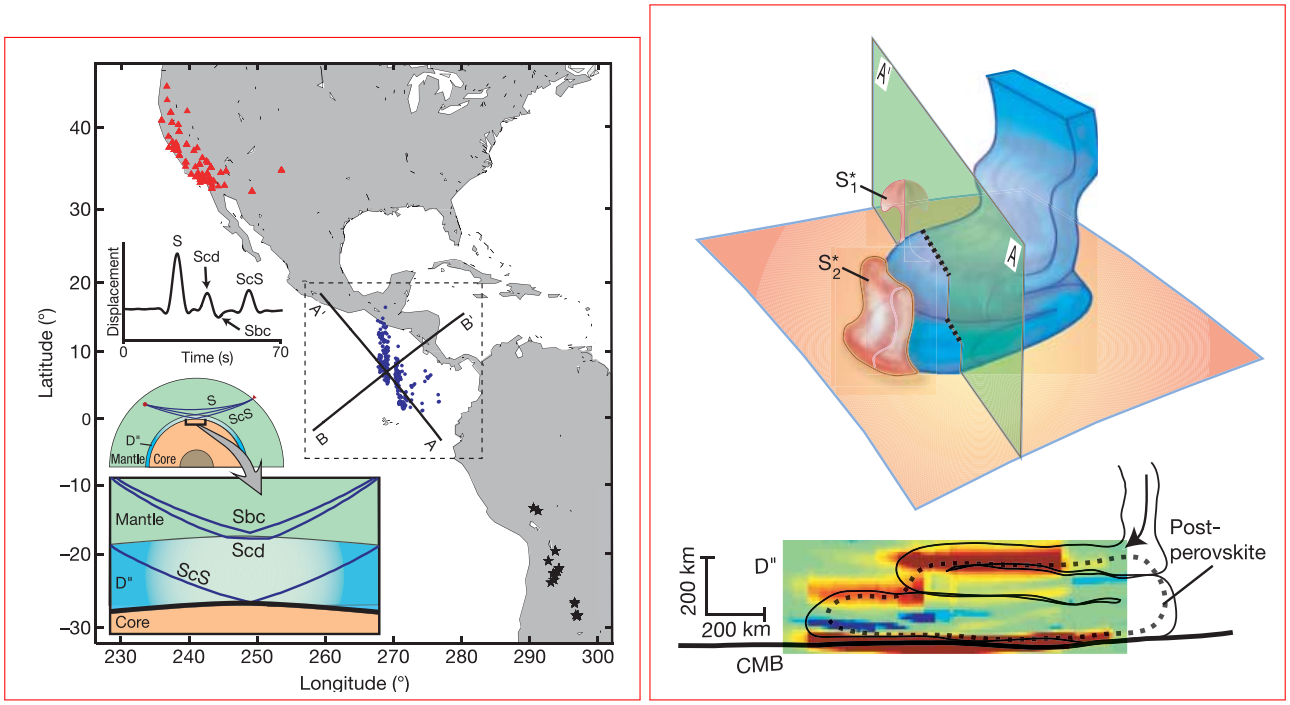


Figure 2: PUBLISHED REFERENCE ONLY: manuscript Figures 1 and 4. Figure 4 includes an interpretive slab schematic. These red-bordered published panels are visually distinct from the three independently generated dataset series.

Set 2: historical

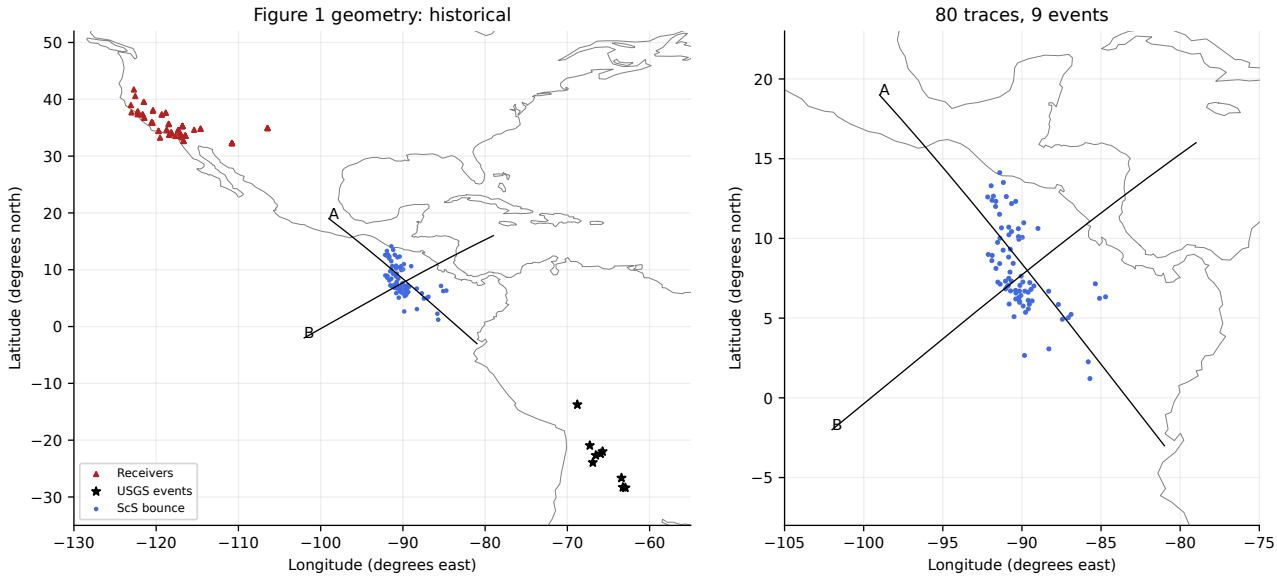


Figure 3: historical; manuscript Figure 1 counterpart: Independently retrieved event and station geometry and calculated ScS bounce points. The two selected section directions are shown. Coastlines are Natural Earth geographic context.

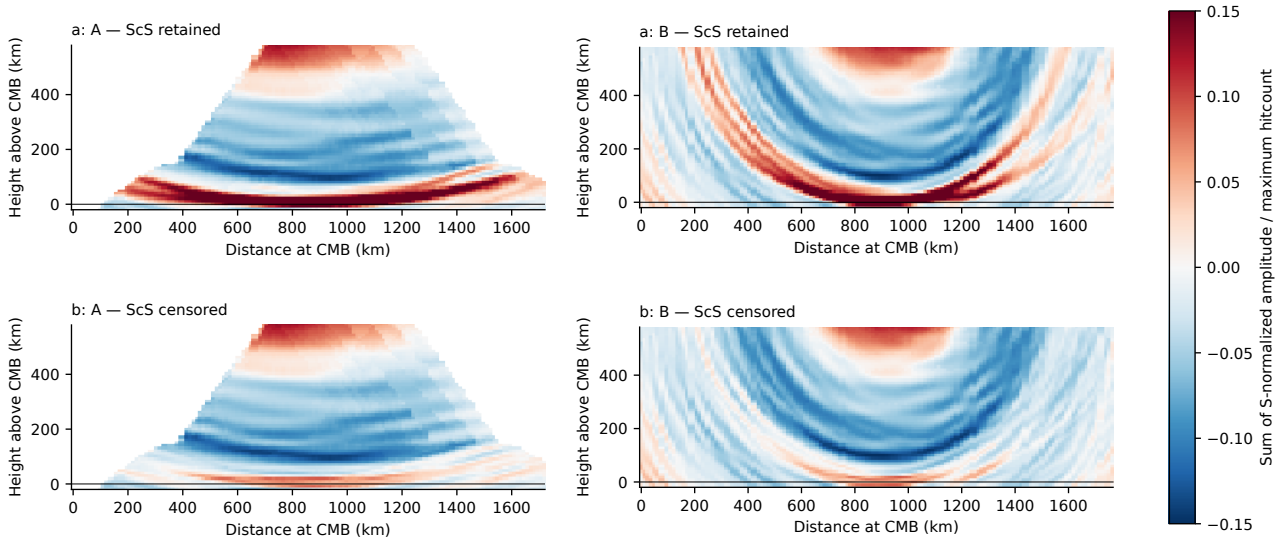


Figure 4: historical; manuscript Figure 2 counterpart: Independent PREM migration. Top: ScS retained; bottom: ScS censored. Left: profile A; right: profile B. Both use ScS as reference and suppress S. Colour is the signed S-normalized stack weighted by relative hitcount, saturated at plus/minus 0.15. White regions fail the 20-trace/three-event display rule.

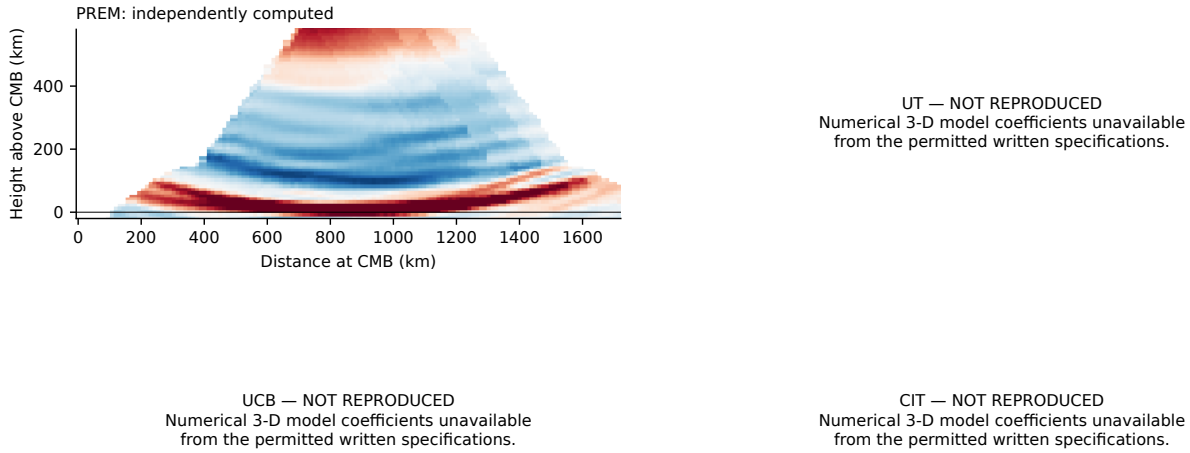


Figure 5: historical; manuscript Figure 3 counterpart: Velocity-model comparison attempt. Only PREM is calculated. Blank, explicitly labelled UT, UCB and CIT panels identify unreproduced results; they are not filled with surrogate models.

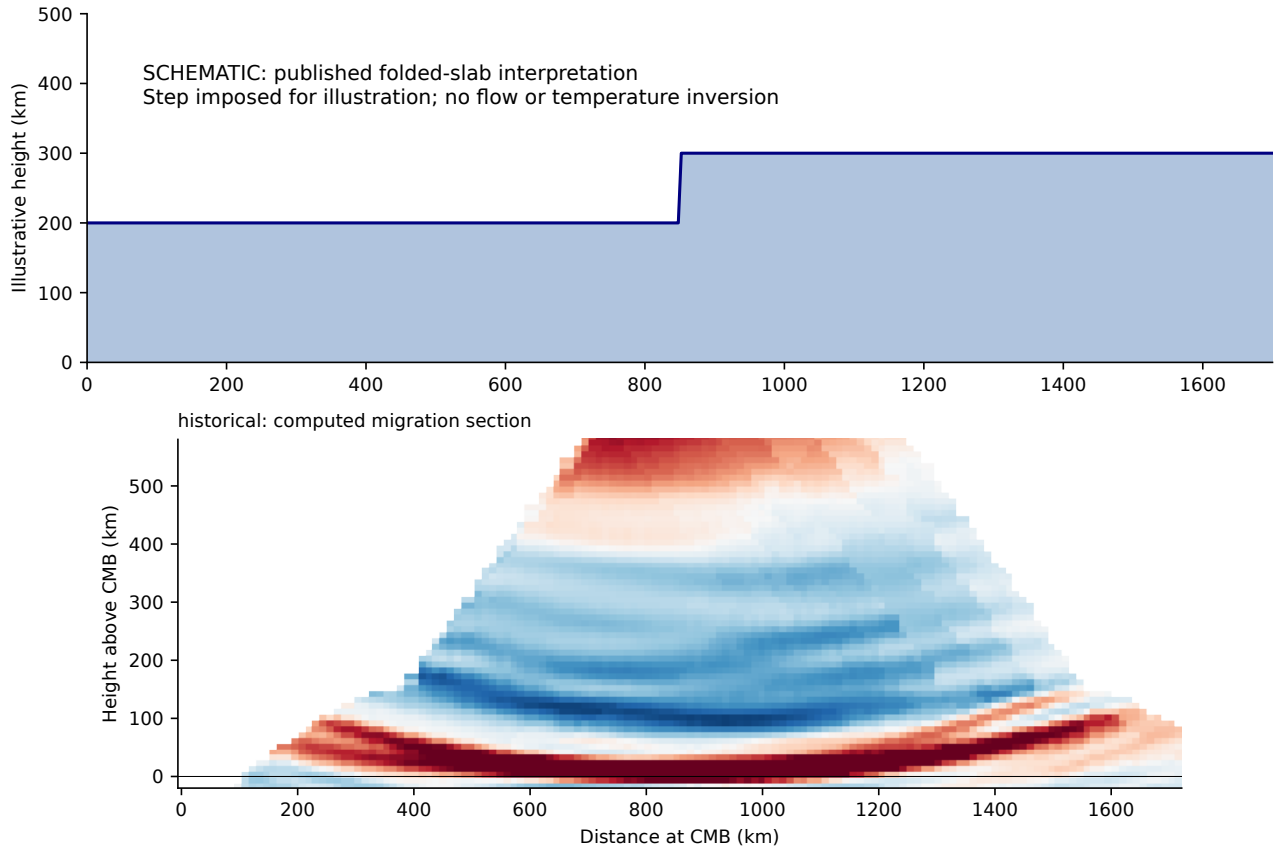


Figure 6: historical; manuscript Figure 4 counterpart: Upper: an explicitly labelled conceptual schematic, with an imposed step and no calculated flow or temperature field. Lower: the independent observational profile A. The schematic is not evidence for a fold.

Set 3: modern_same

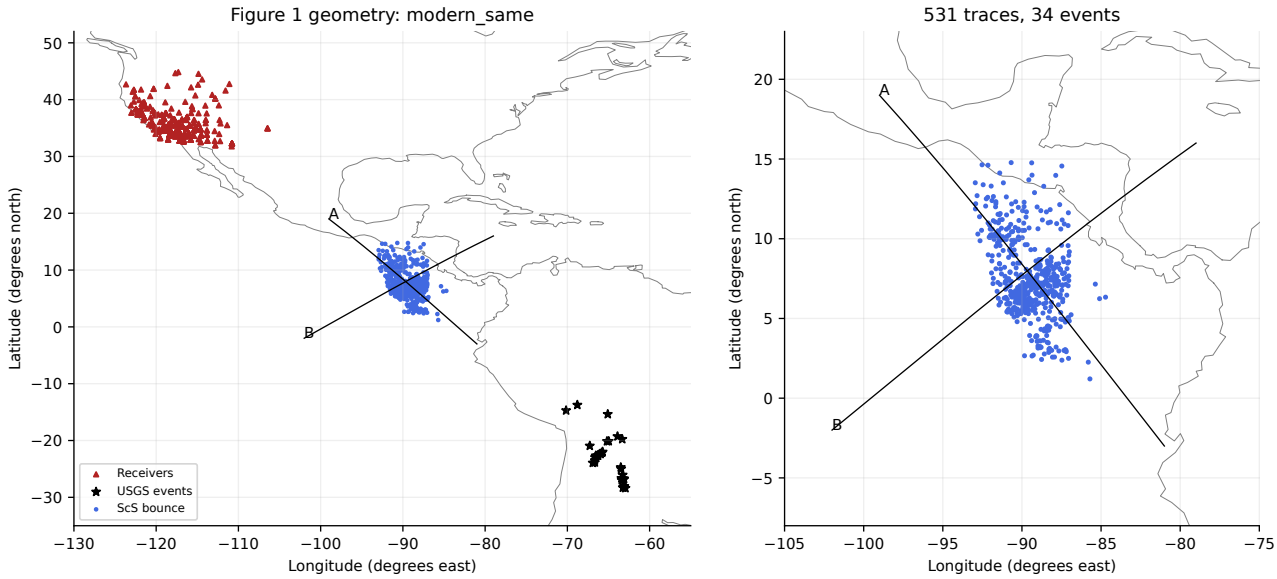


Figure 7: modern_same; manuscript Figure 1 counterpart: Independently retrieved event and station geometry and calculated ScS bounce points. The two selected section directions are shown. Coastlines are Natural Earth geographic context.

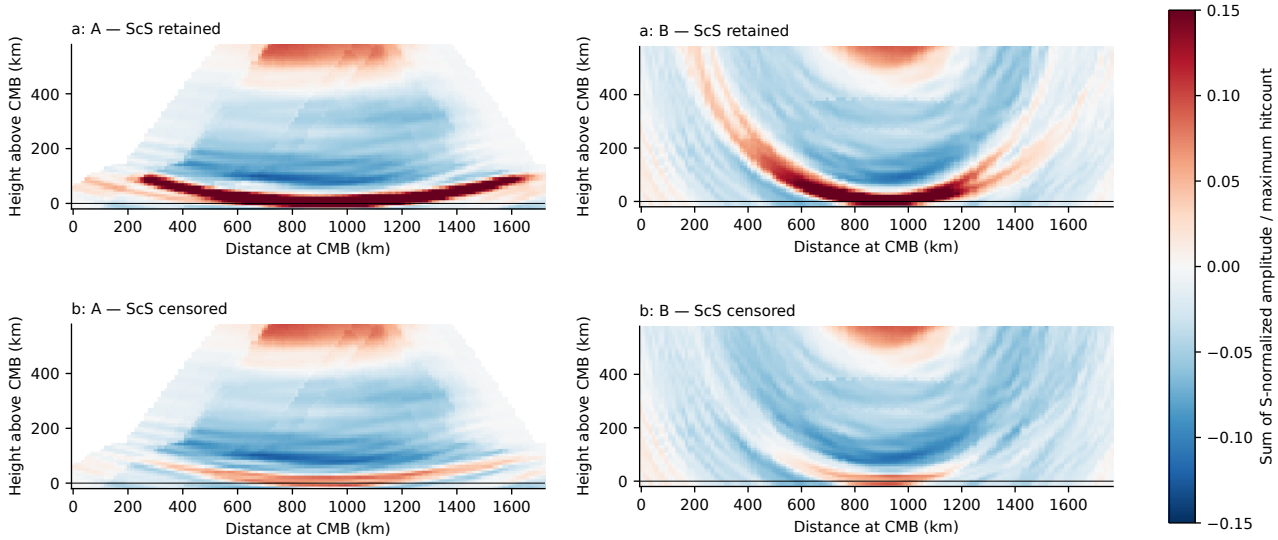


Figure 8: modern_same; manuscript Figure 2 counterpart: Independent PREM migration. Top: ScS retained; bottom: ScS censored. Left: profile A; right: profile B. Both use ScS as reference and suppress S. Colour is the signed S-normalized stack weighted by relative hitcount, saturated at plus/minus 0.15. White regions fail the 20-trace/three-event display rule.

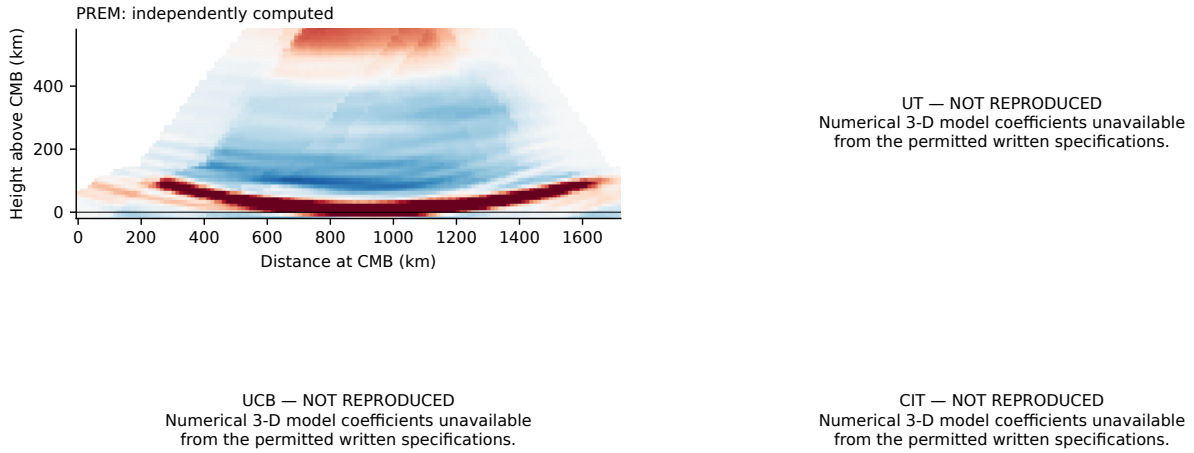


Figure 9: modern_same; manuscript Figure 3 counterpart: Velocity-model comparison attempt. Only PREM is calculated. Blank, explicitly labelled UT, UCB and CIT panels identify unreproduced results; they are not filled with surrogate models.

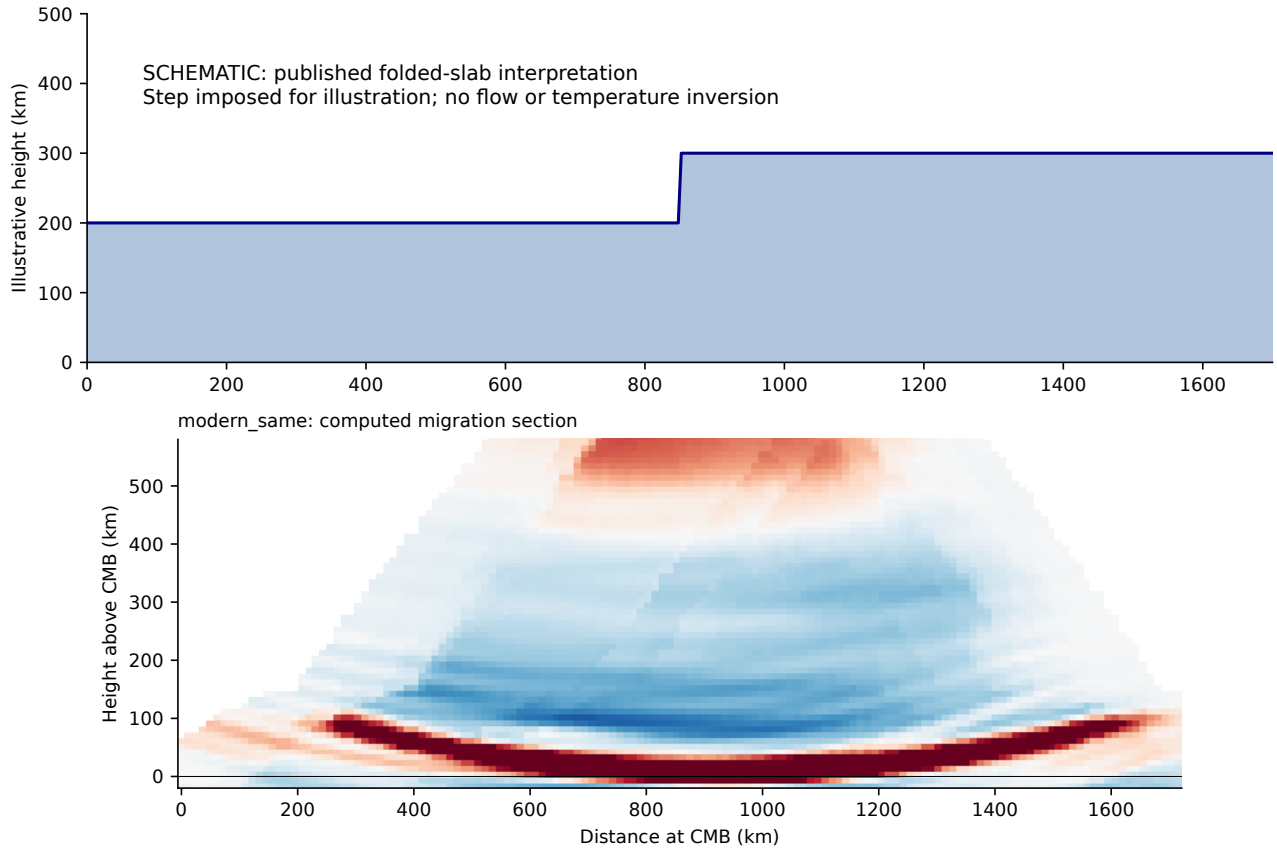


Figure 10: modern_same; manuscript Figure 4 counterpart: Upper: an explicitly labelled conceptual schematic, with an imposed step and no calculated flow or temperature field. Lower: the independent observational profile A. The schematic is not evidence for a fold.

Set 4: expanded

The wider calculation covers 10 degrees S–25 degrees N and 110–70 degrees W. Profiles A and B are retained for direct comparison; the horizontal slices in supplementary Figure 5 show the additional imaged region, with the same coverage threshold.

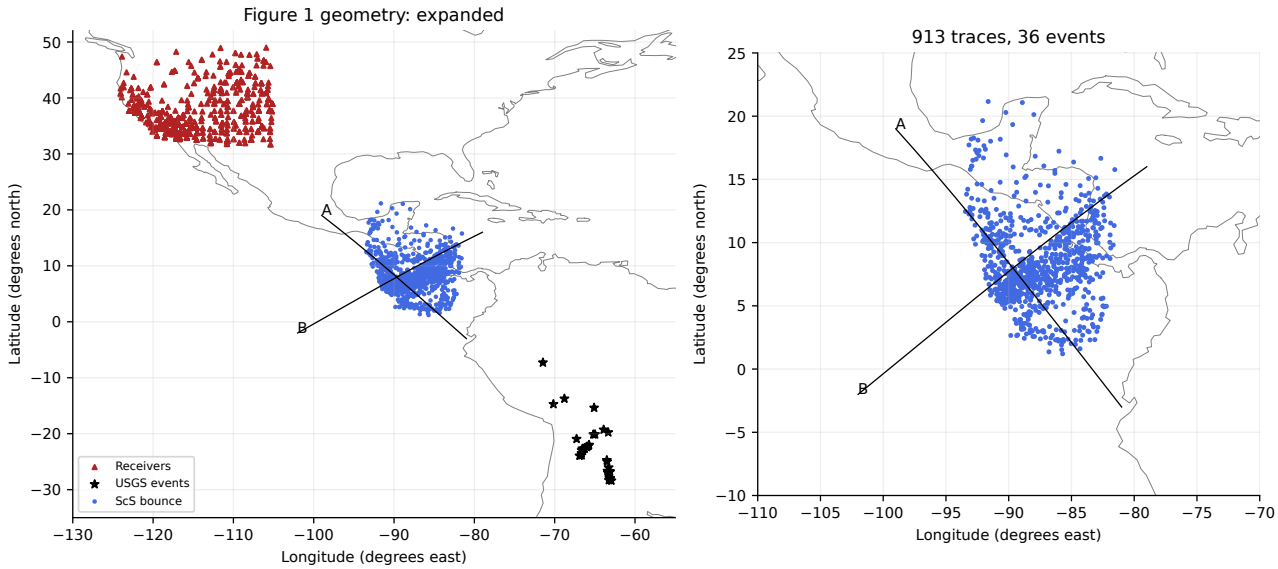


Figure 11: expanded; manuscript Figure 1 counterpart: Independently retrieved event and station geometry and calculated ScS bounce points. The two selected section directions are shown. Coastlines are Natural Earth geographic context.

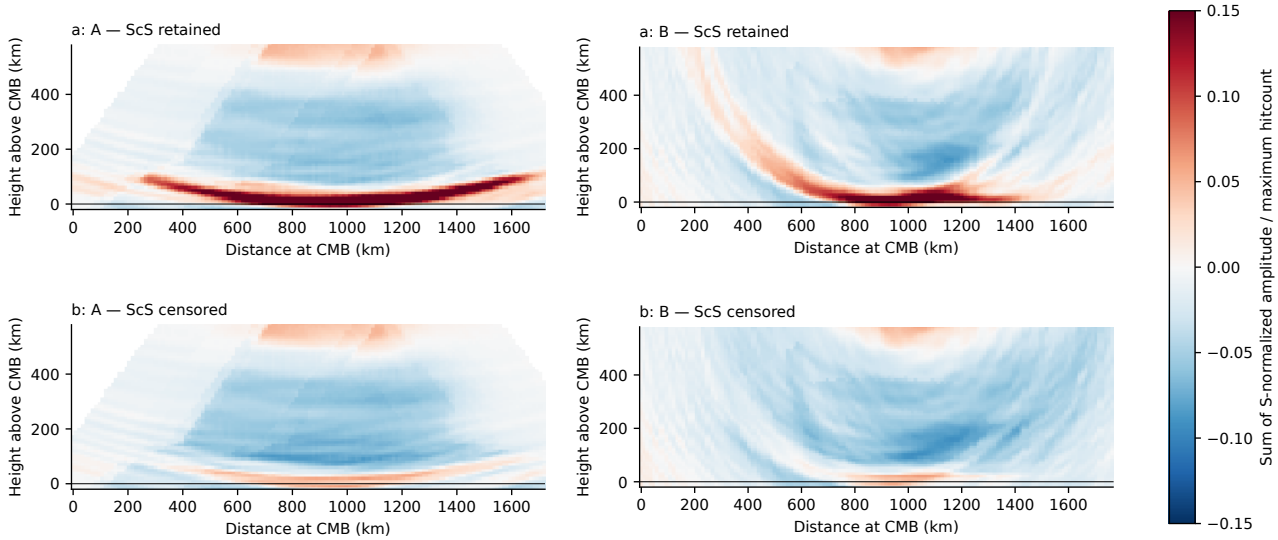


Figure 12: expanded; manuscript Figure 2 counterpart: Independent PREM migration. Top: ScS retained; bottom: ScS censored. Left: profile A; right: profile B. Both use ScS as reference and suppress S. Colour is the signed S-normalized stack weighted by relative hitcount, saturated at plus/minus 0.15. White regions fail the 20-trace/three-event display rule.

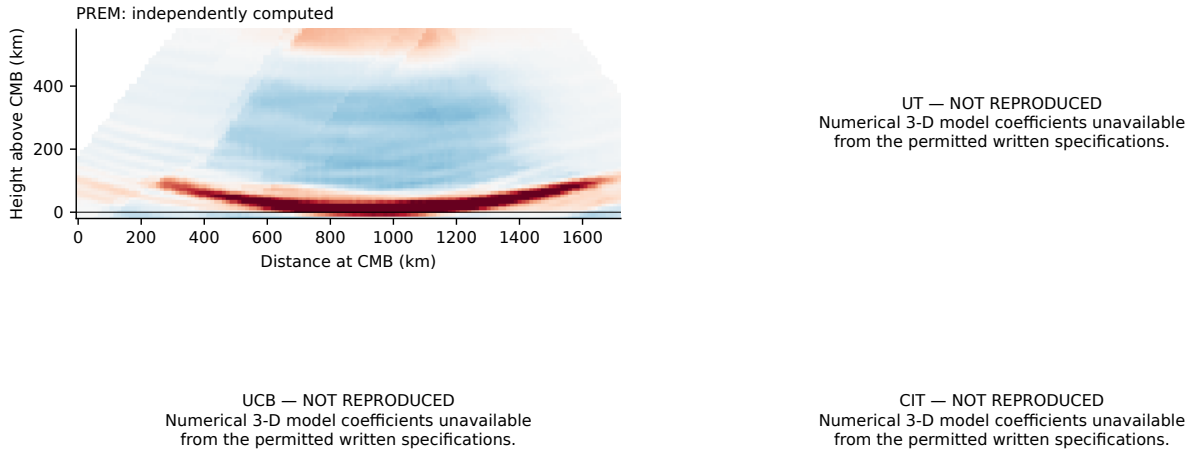


Figure 13: expanded; manuscript Figure 3 counterpart: Velocity-model comparison attempt. Only PREM is calculated. Blank, explicitly labelled UT, UCB and CIT panels identify unreproduced results; they are not filled with surrogate models.

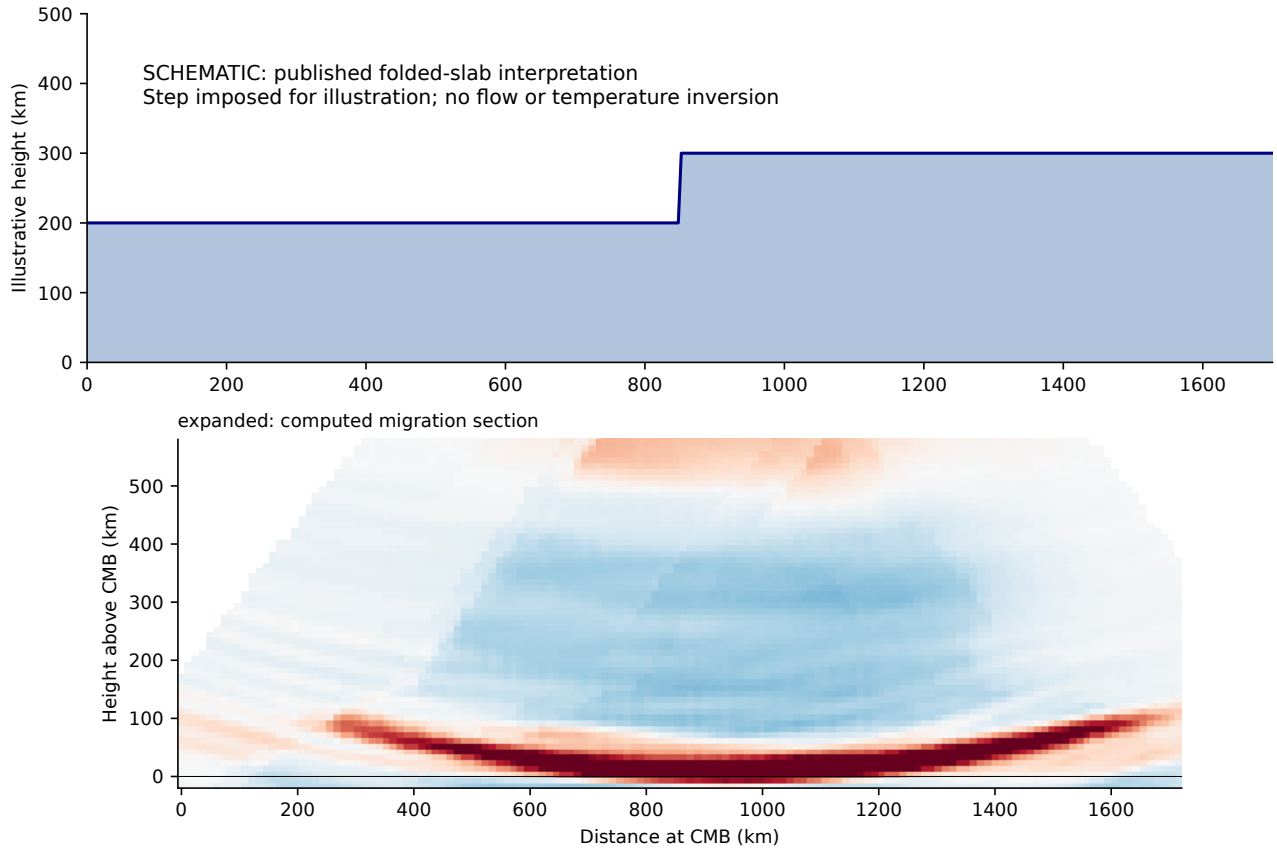


Figure 14: expanded; manuscript Figure 4 counterpart: Upper: an explicitly labelled conceptual schematic, with an imposed step and no calculated flow or temperature field. Lower: the independent observational profile A. The schematic is not evidence for a fold.

5 Supplementary calculations

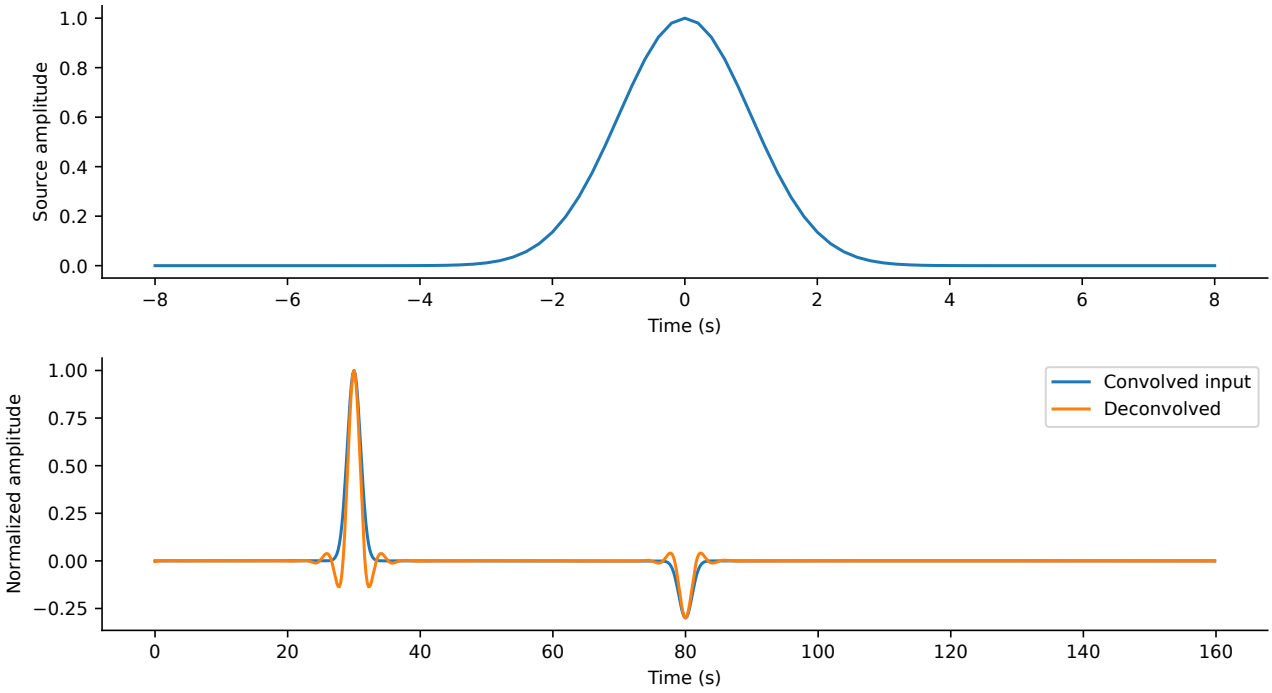


Figure 15: Synthetic deconvolution consistency test. The input consists of two known signed impulses convolved with the displayed wavelet. This is a validation signal, not Earth data.

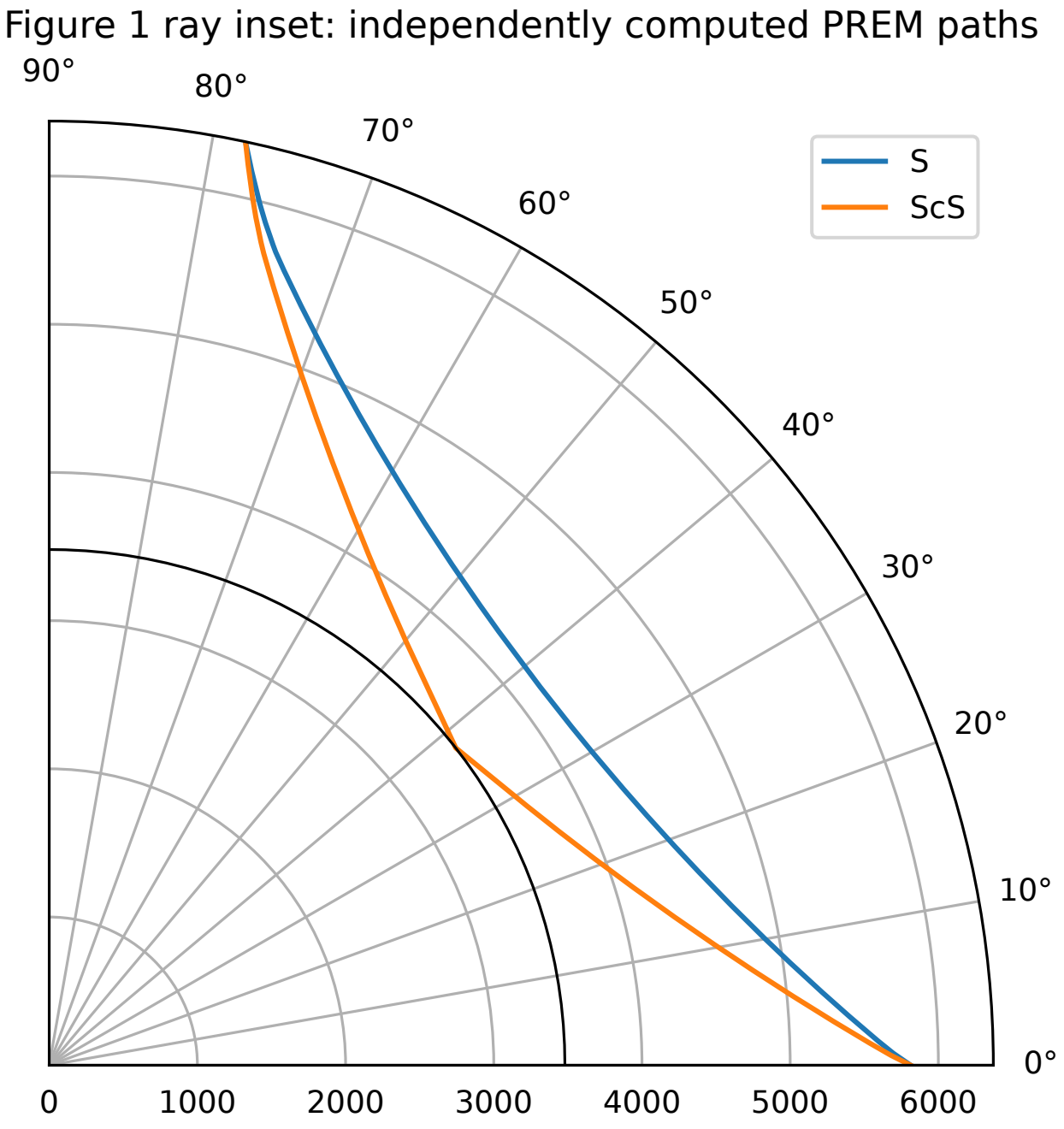


Figure 16: Independently calculated S and ScS paths for a 558 km source at 78 degrees. This is a numerical ray diagram; the publication does not specify the exact illustrative source used in its inset.

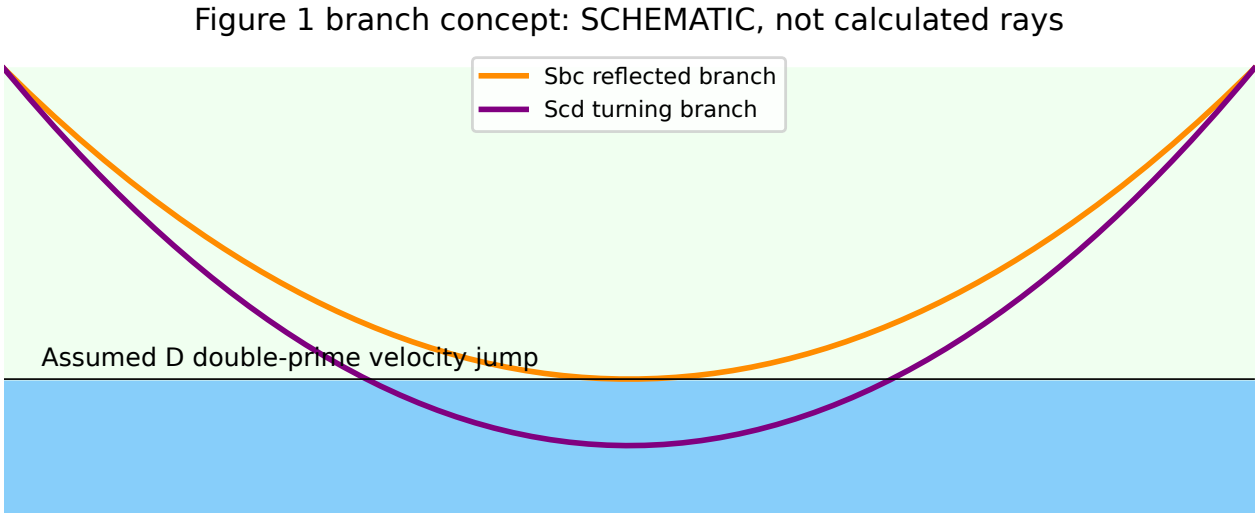


Figure 17: Redrawn conceptual companion to the Figure 1 triplication inset. The curves only distinguish reflection above the jump from turning below it. They do not satisfy a calculated ray equation and are not reproduced numerical results.

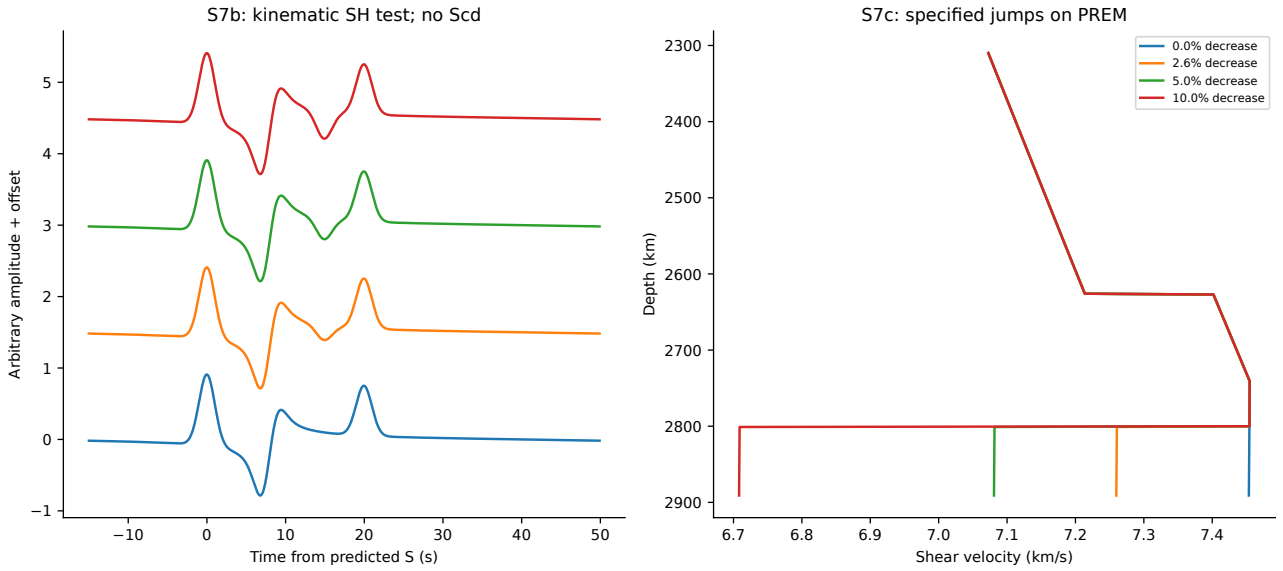


Figure 18: Partial attempt at supplementary Figure 7b,c: local SH reflection coefficients and explicitly assumed layered velocity profiles. There is no refracted Scd branch or full-wave reflectivity calculation; these curves cannot reproduce the published triplication amplitudes.

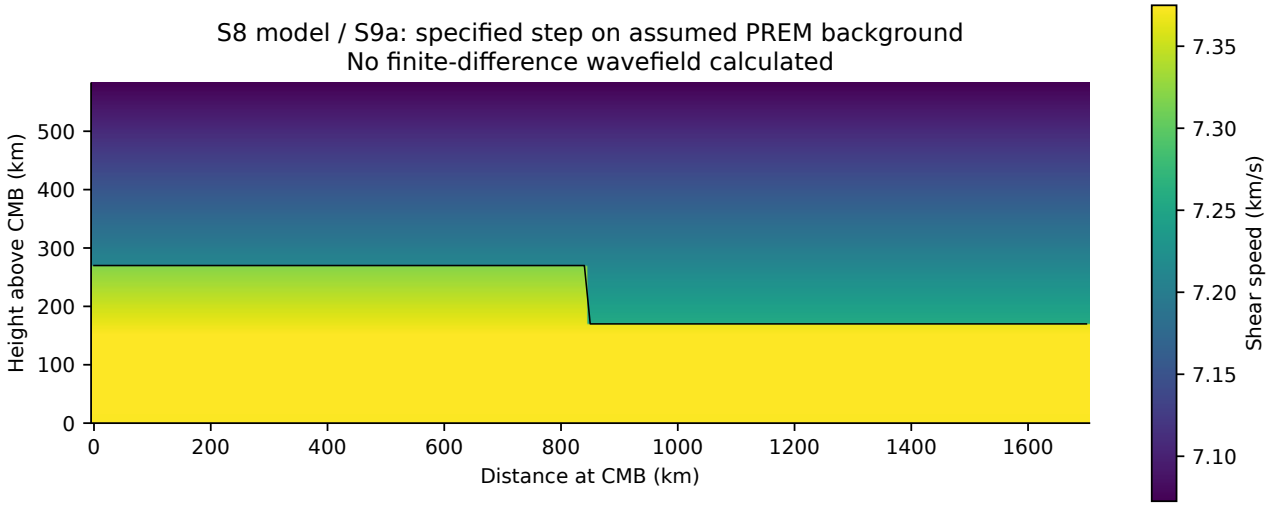


Figure 19: Numerical input-model illustration for supplementary Figures 8 and 9a. The stated 170/270 km heights and 1.5 percent jump are imposed on PREM. No full-wave propagation has been performed.

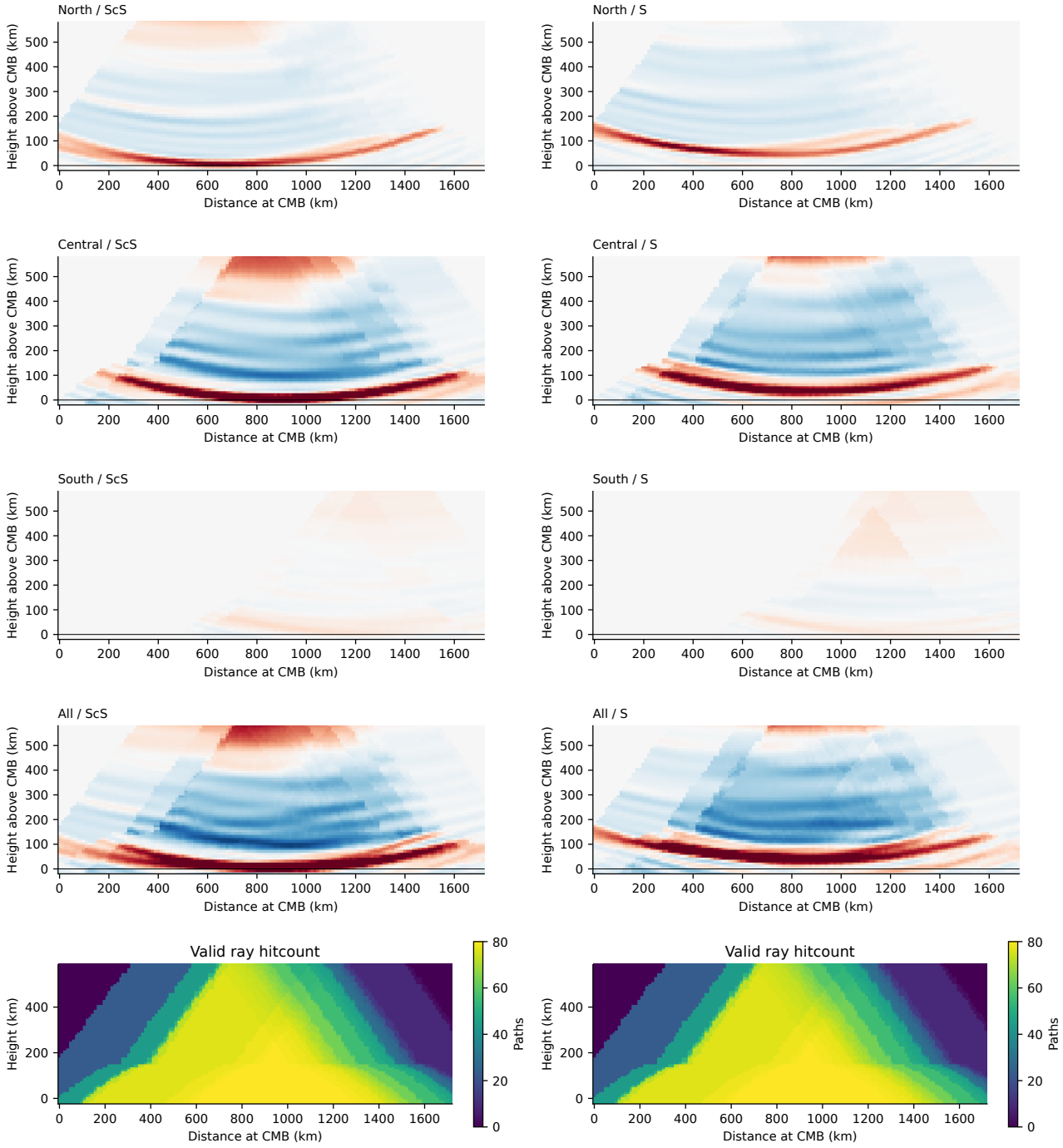


Figure 20: historical; independent supplementary figure 2 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

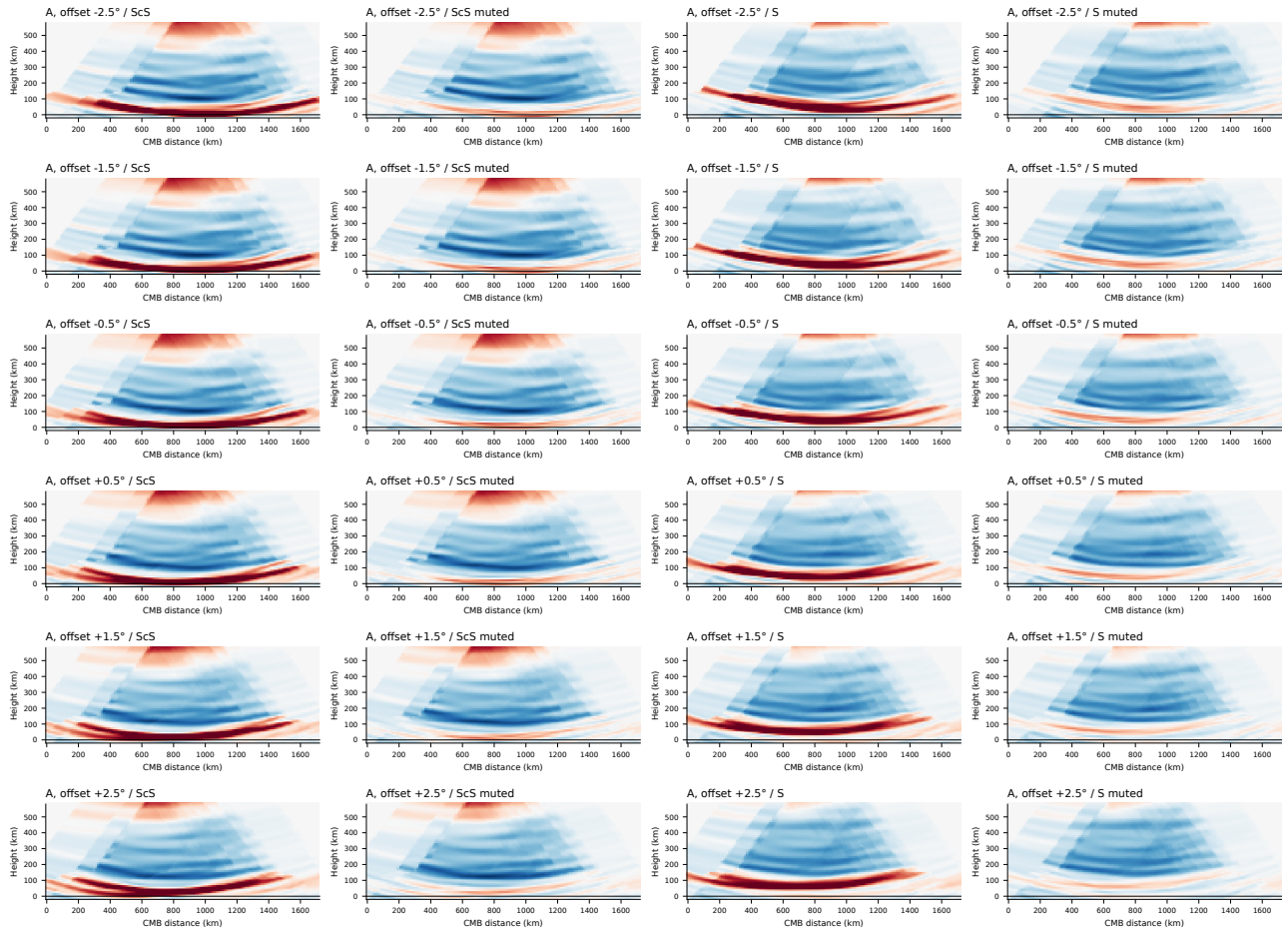


Figure 21: historical; independent supplementary figure 3a counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

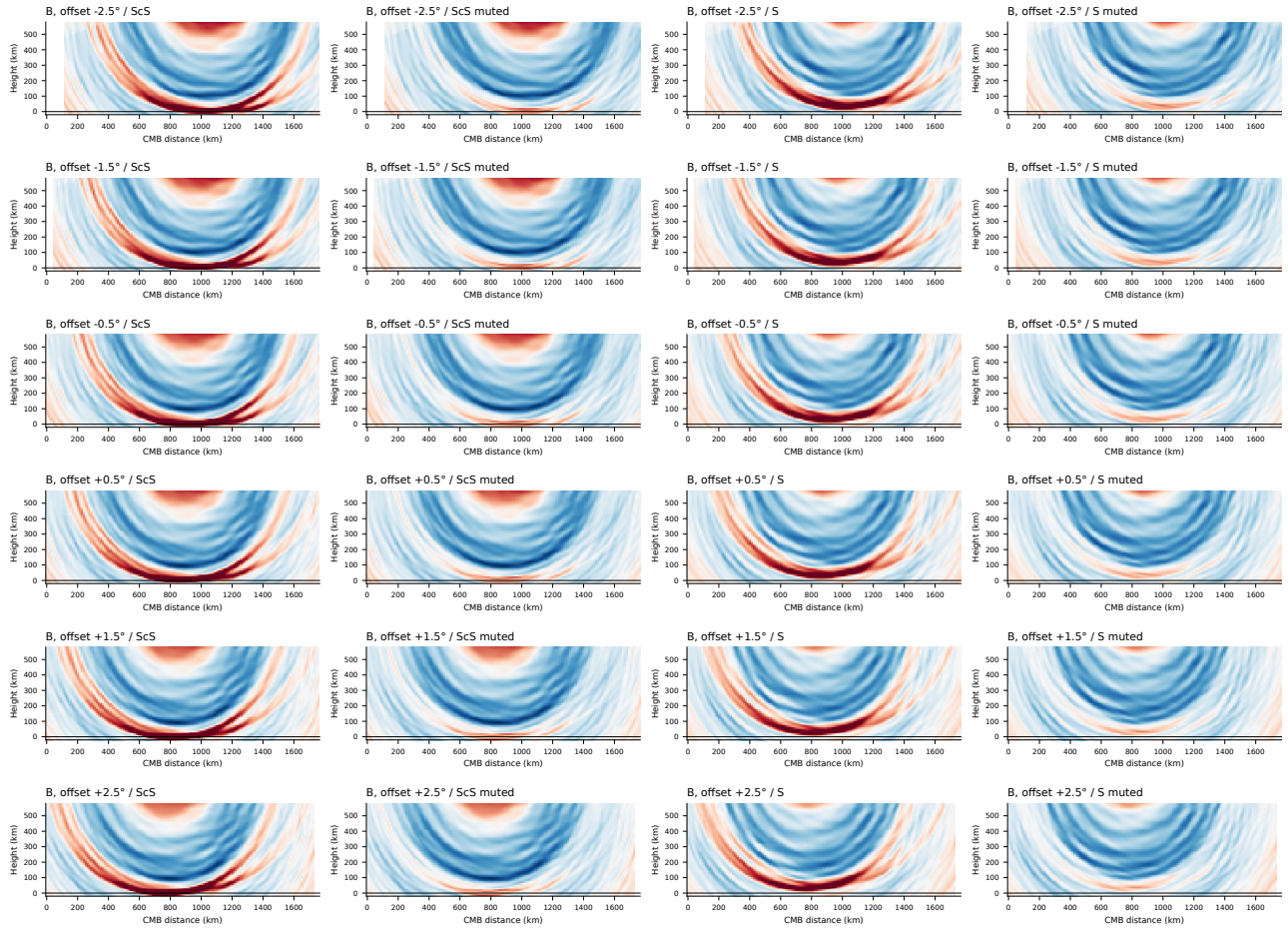


Figure 22: historical; independent supplementary figure 3b counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

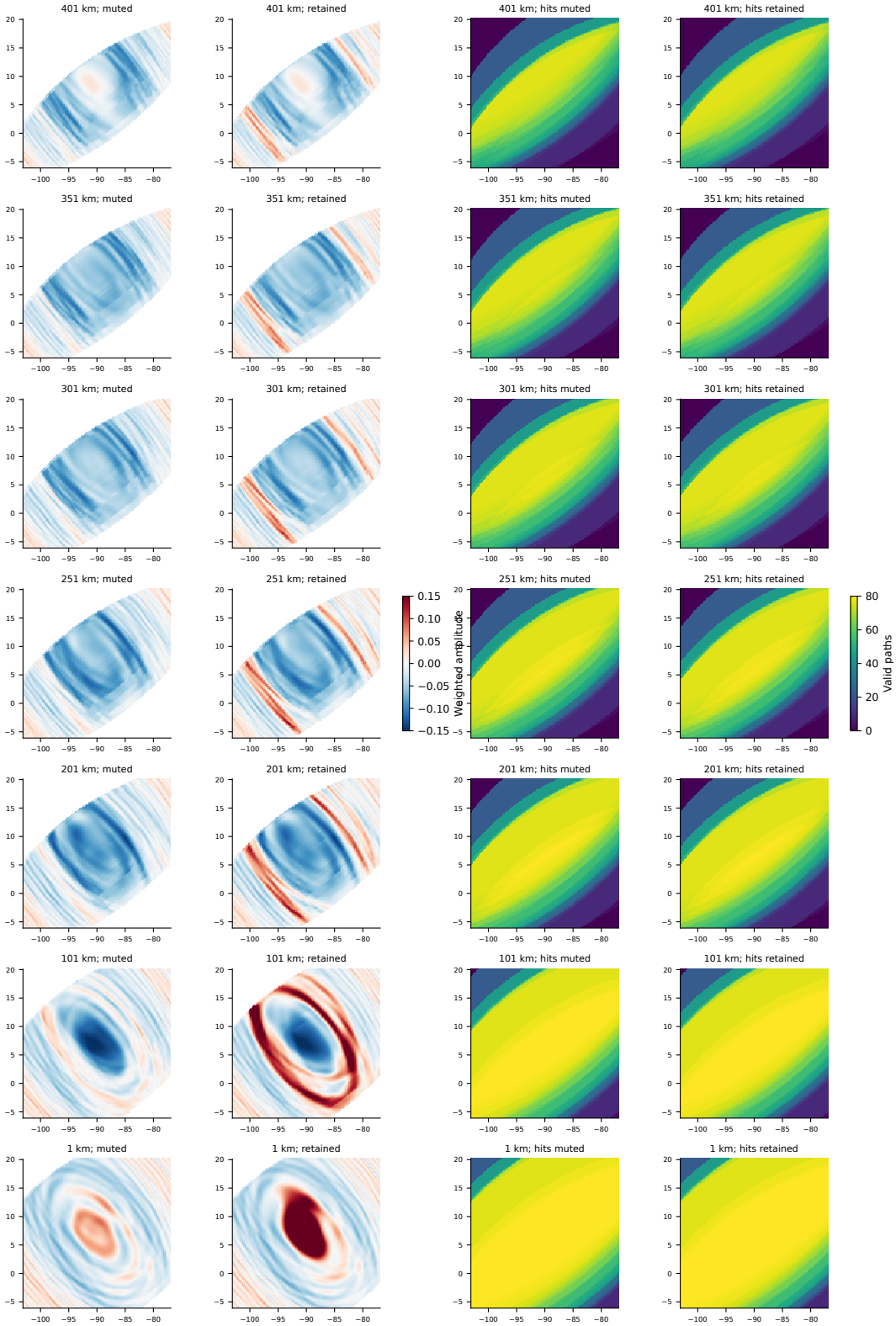


Figure 23: historical; independent supplementary figure 5 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

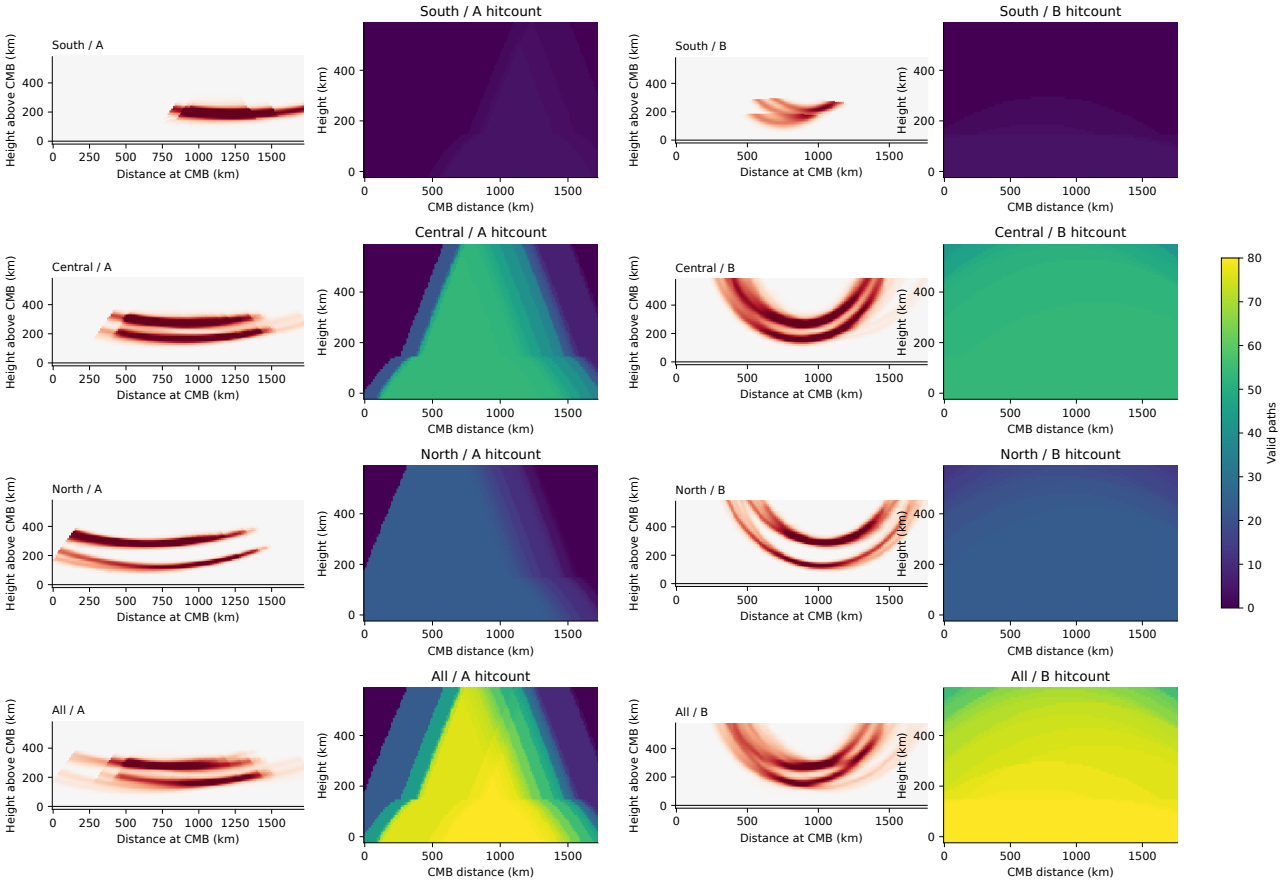


Figure 24: historical; independent supplementary figure 6 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

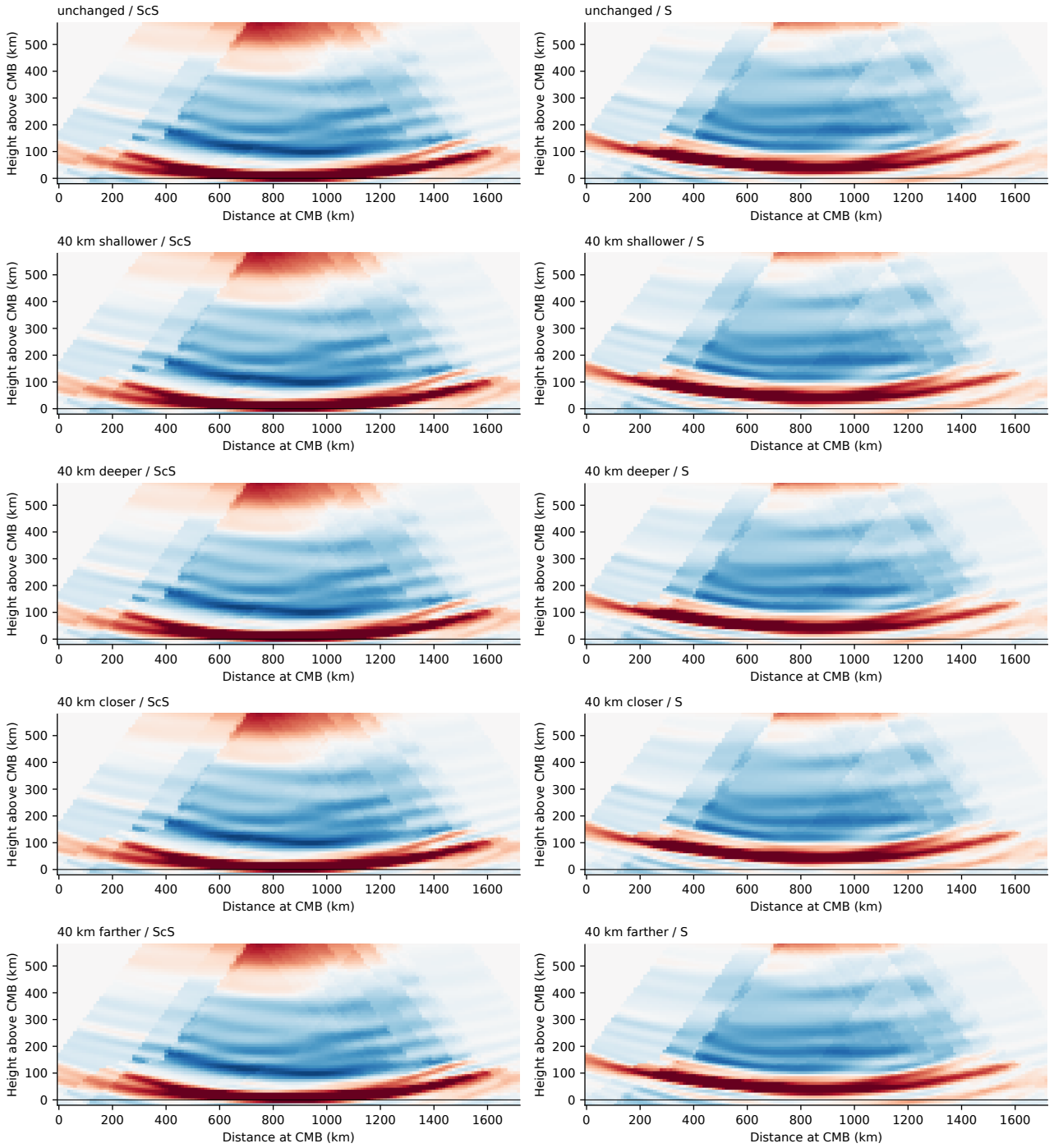


Figure 25: historical; independent supplementary figure 10 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

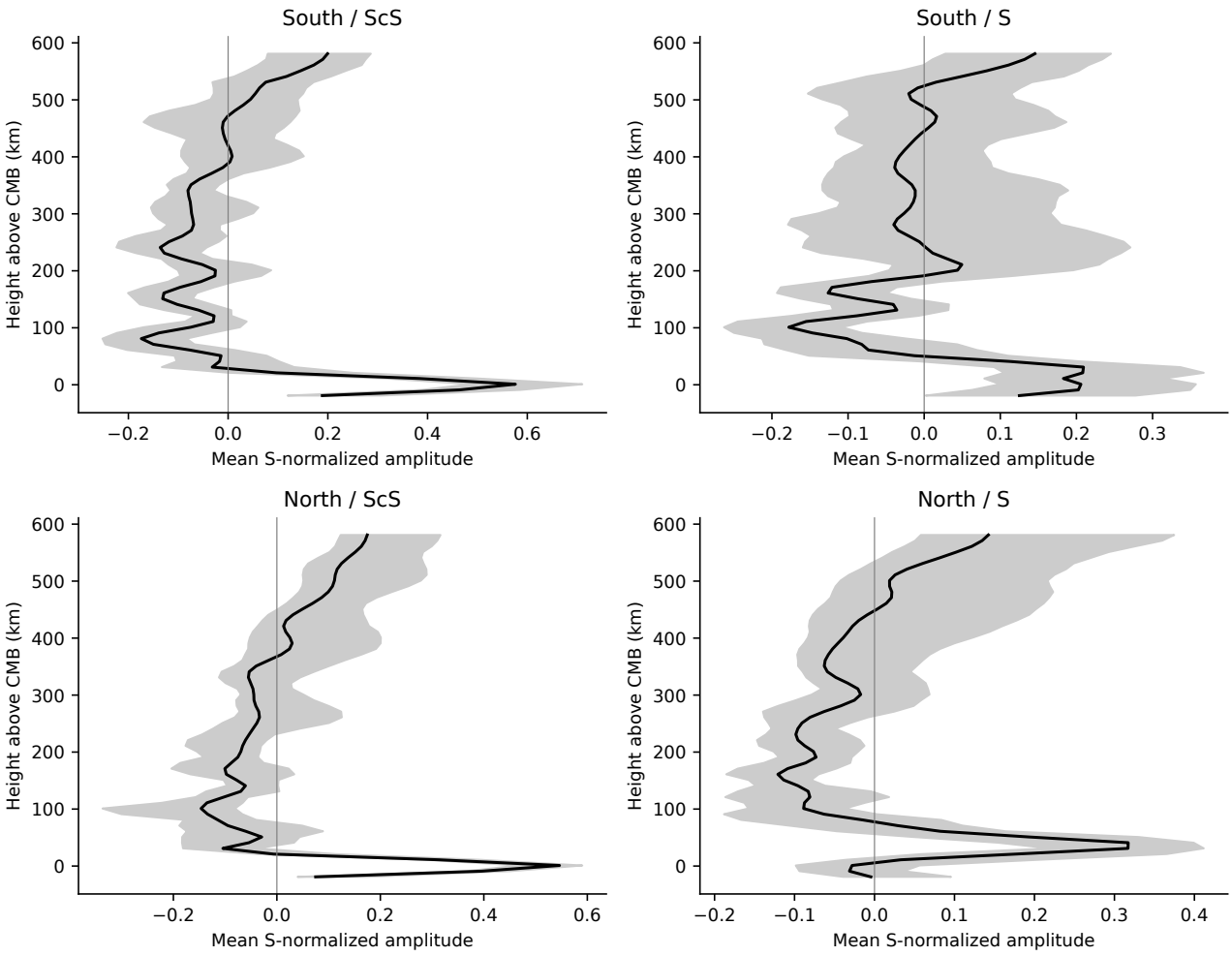


Figure 26: historical; independent supplementary figure 11 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

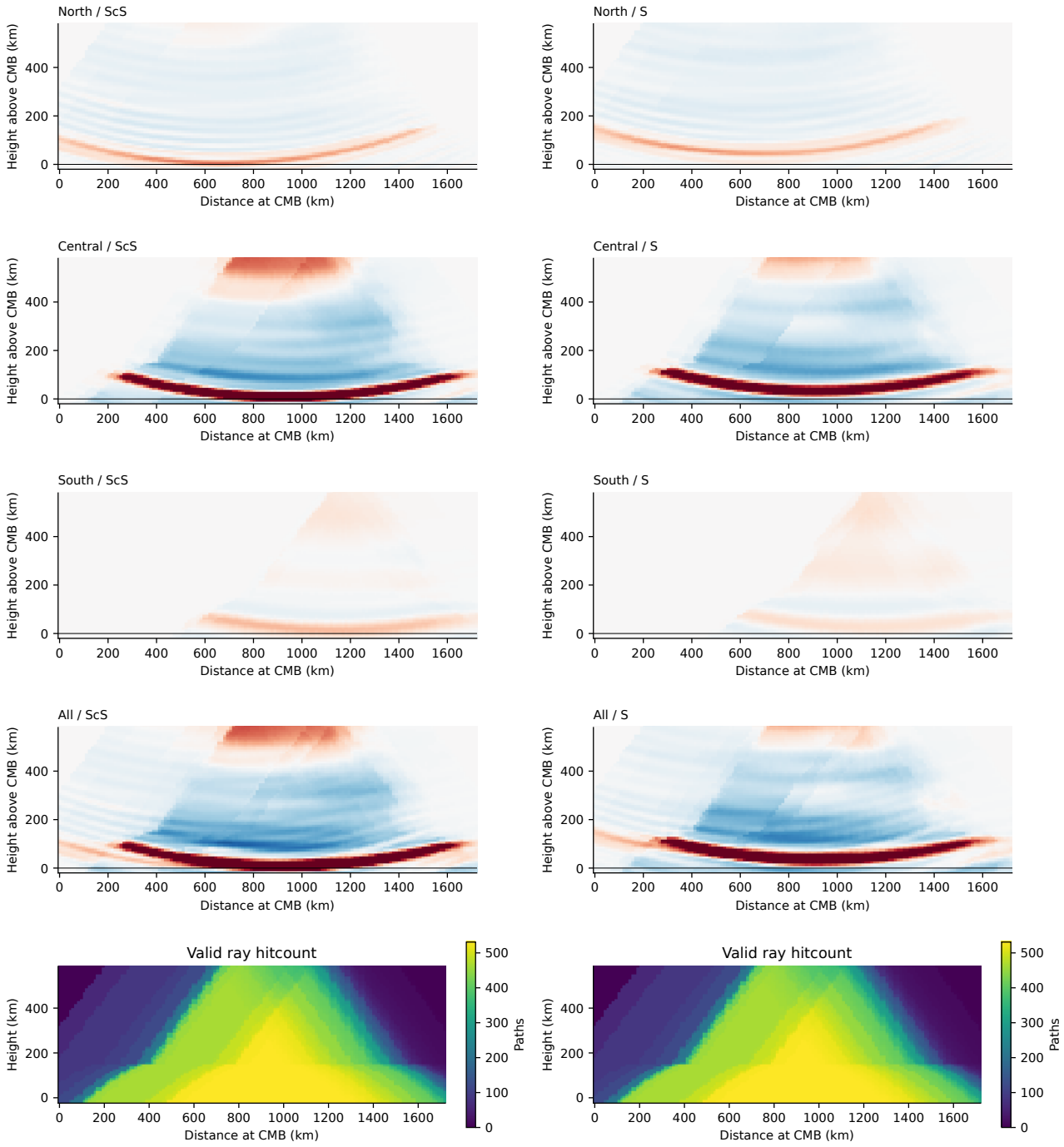


Figure 27: modern_same; independent supplementary figure 2 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

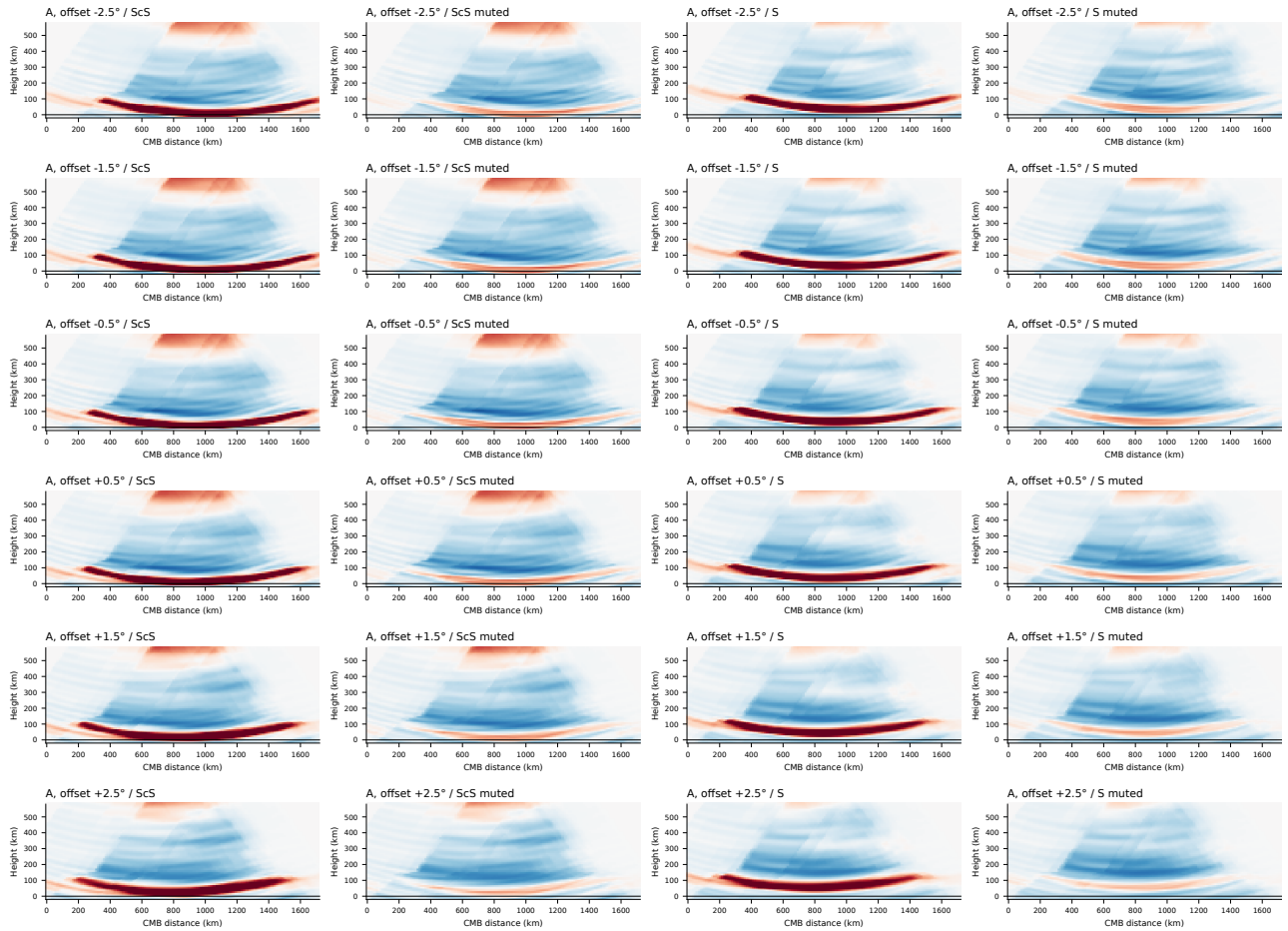


Figure 28: modern_same; independent supplementary figure 3a counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

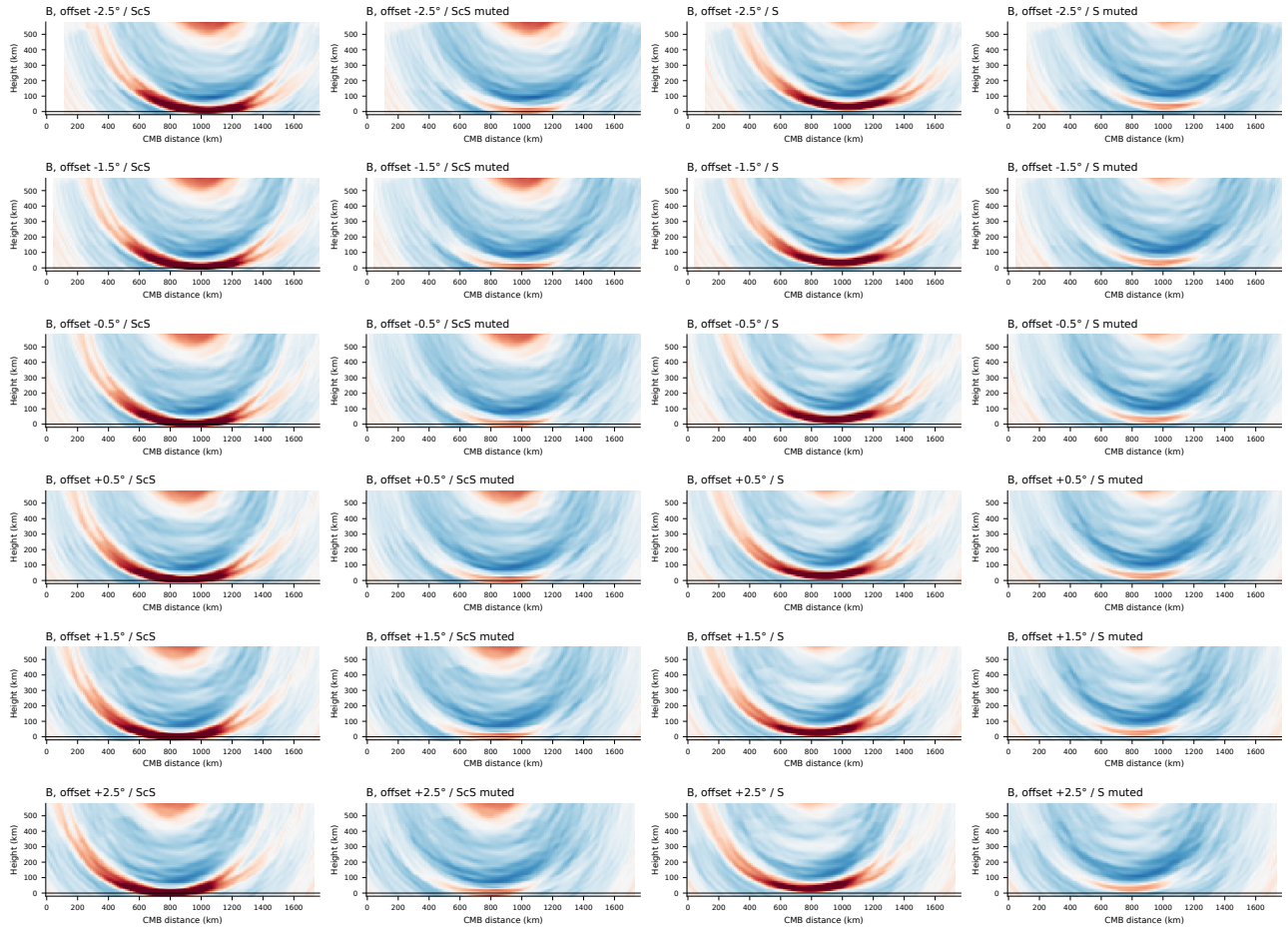


Figure 29: modern_same; independent supplementary figure 3b counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

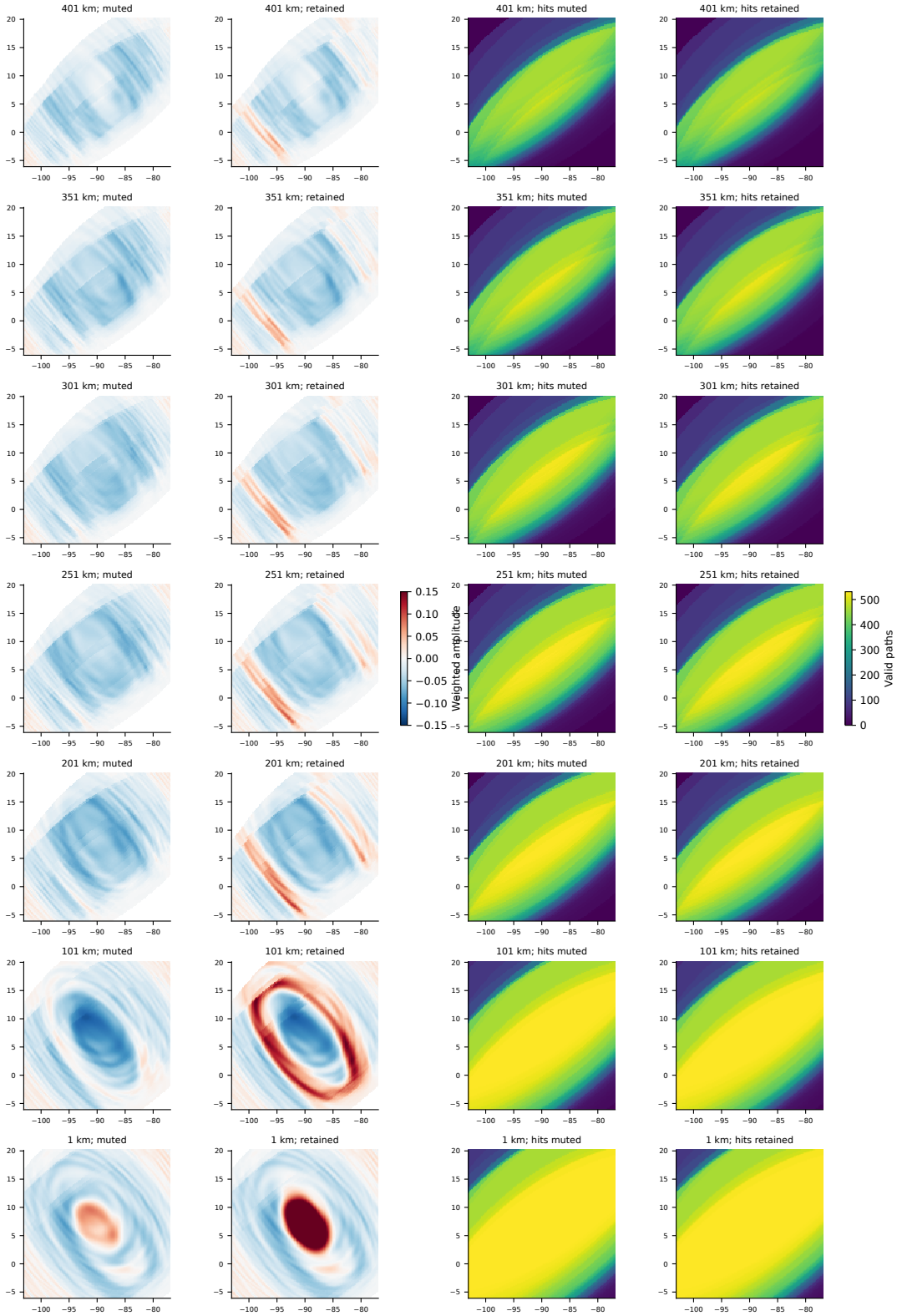


Figure 30: modern_same; independent supplementary figure 5 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

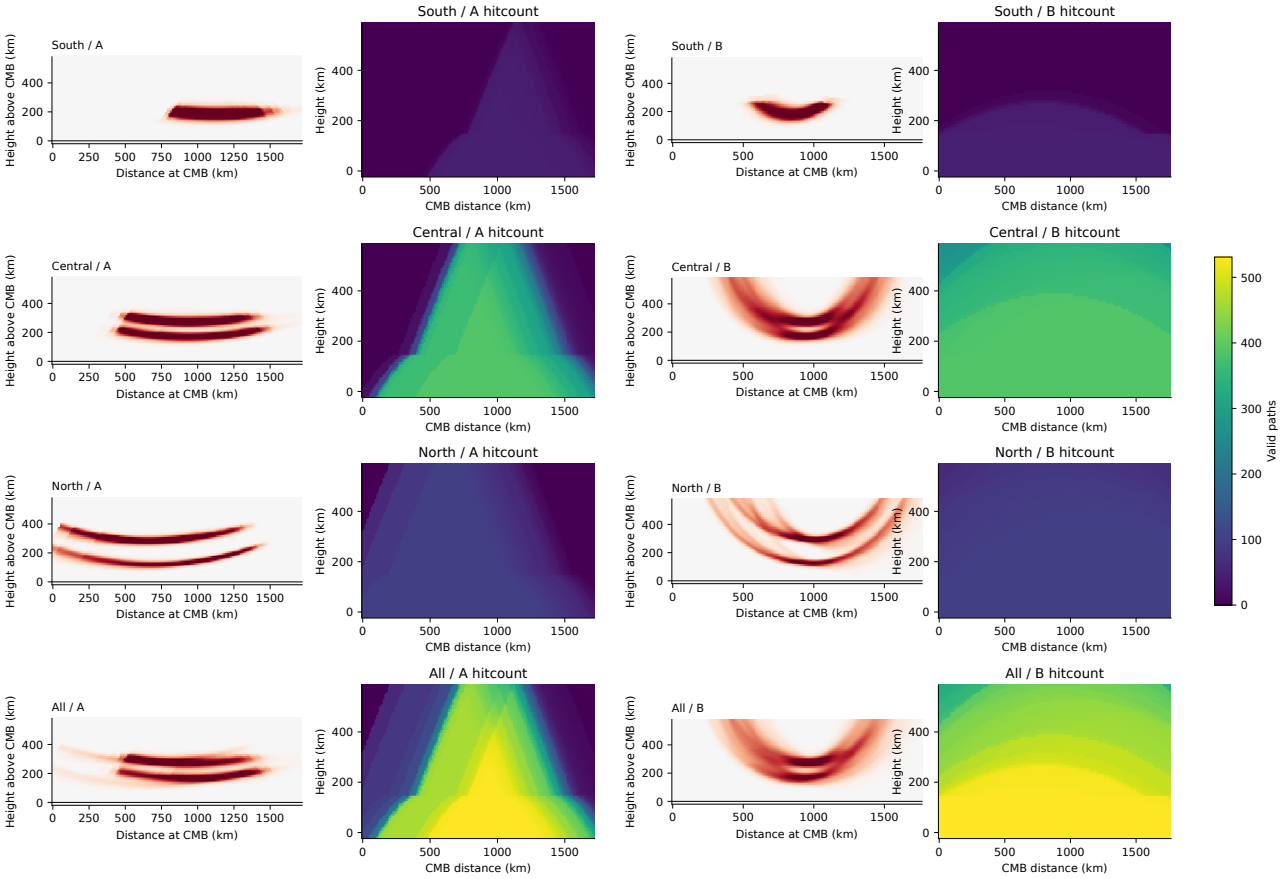


Figure 31: modern_same; independent supplementary figure 6 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

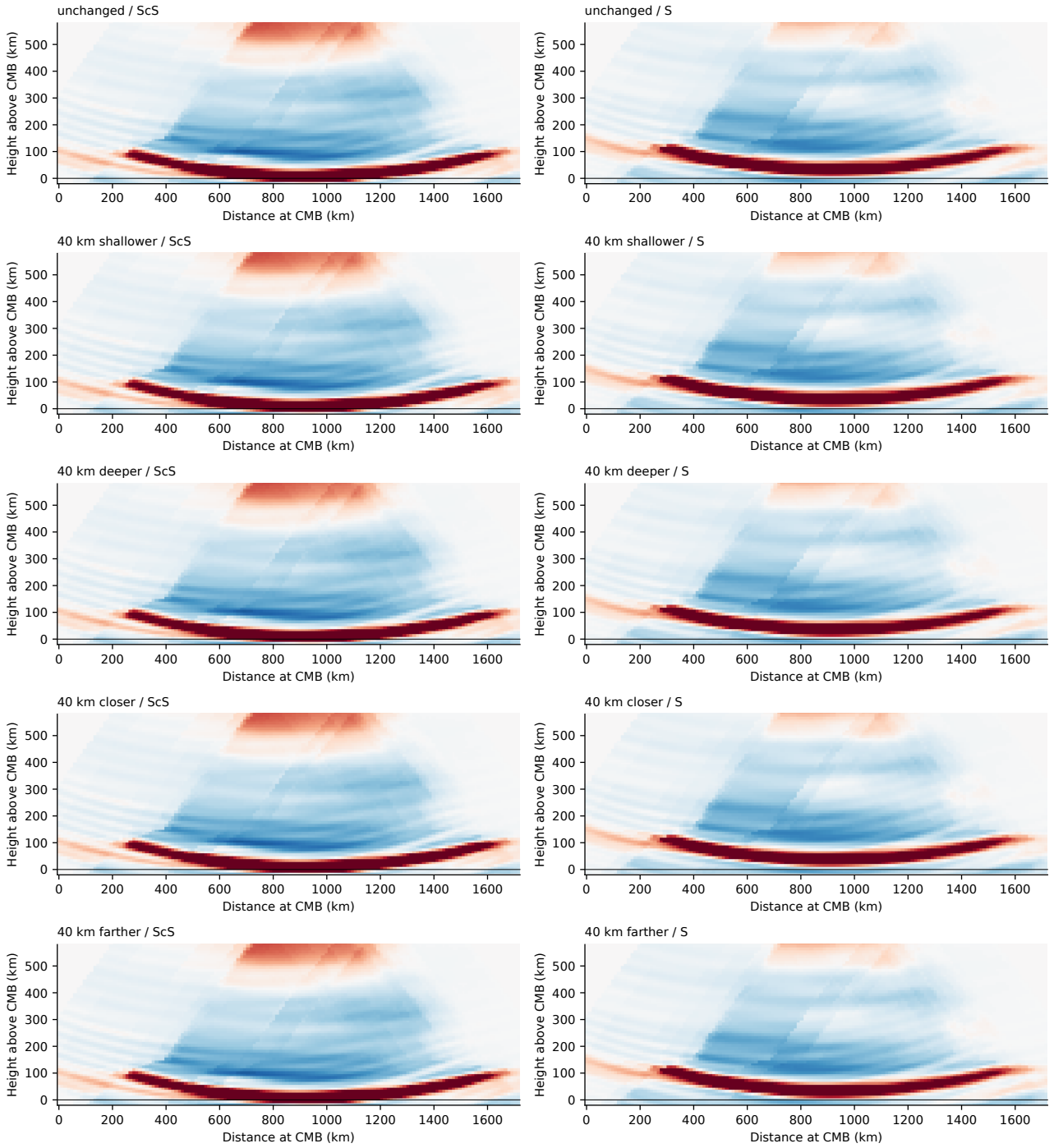


Figure 32: modern_same; independent supplementary figure 10 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

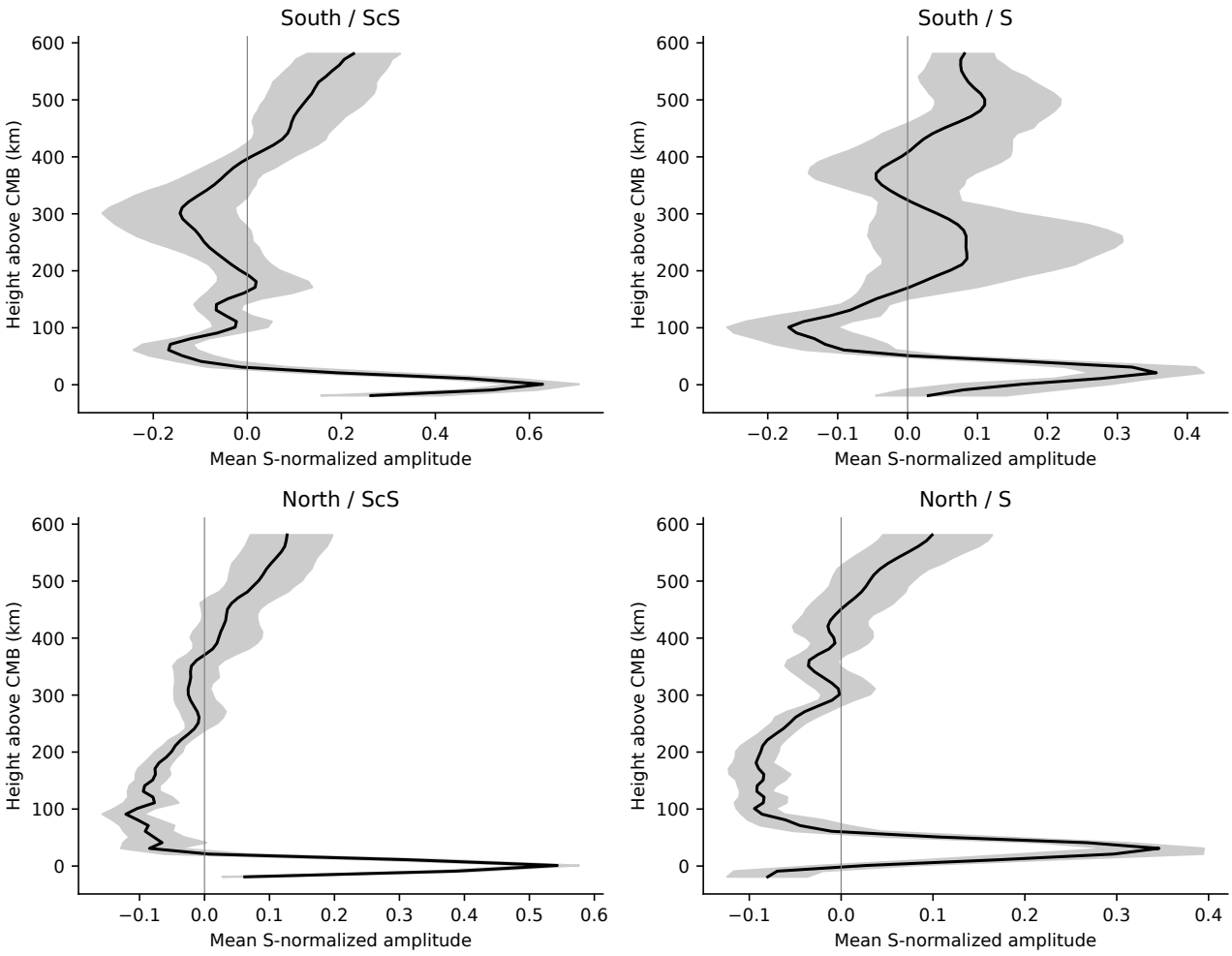


Figure 33: modern_same; independent supplementary figure 11 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

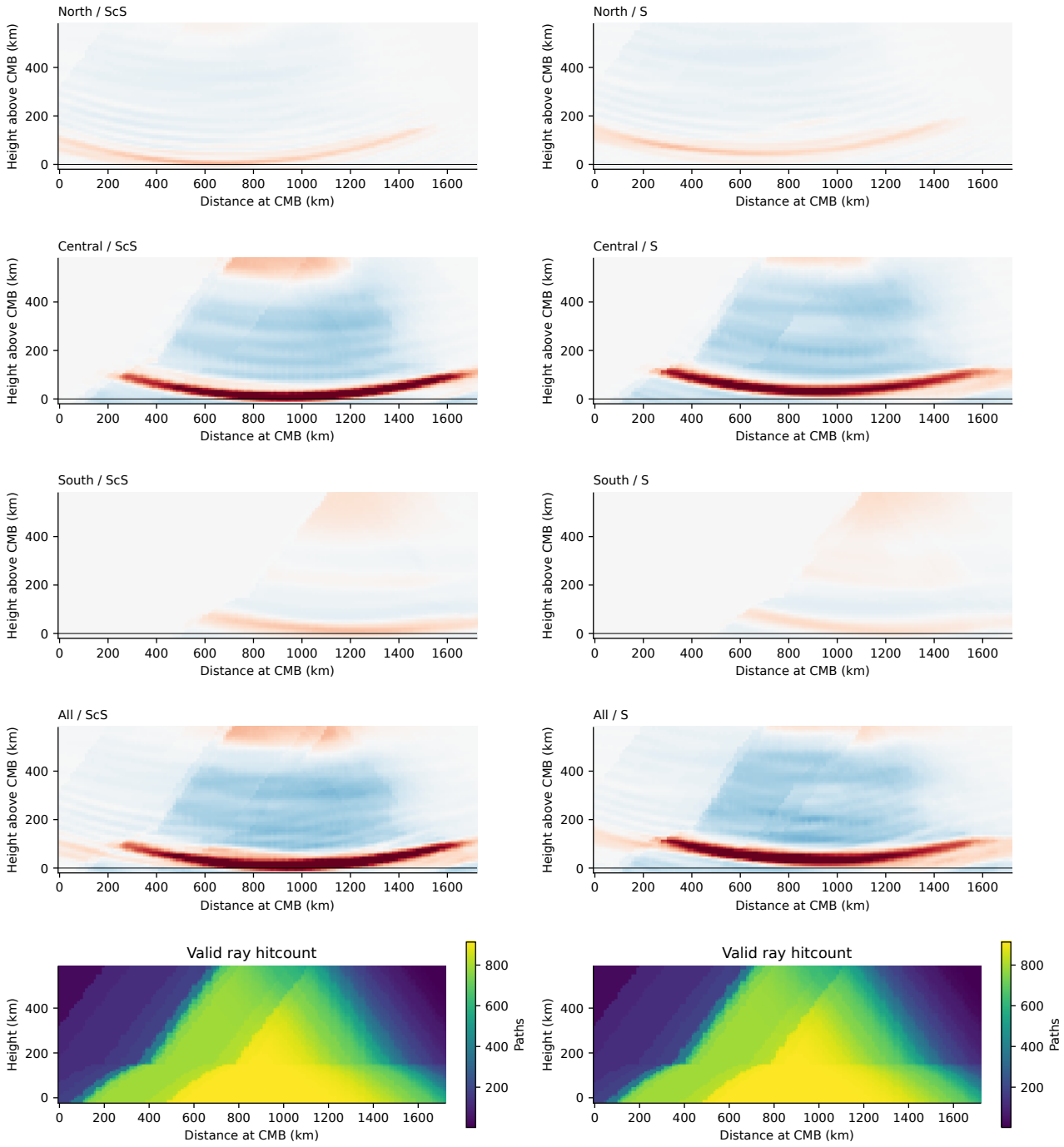


Figure 34: expanded; independent supplementary figure 2 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

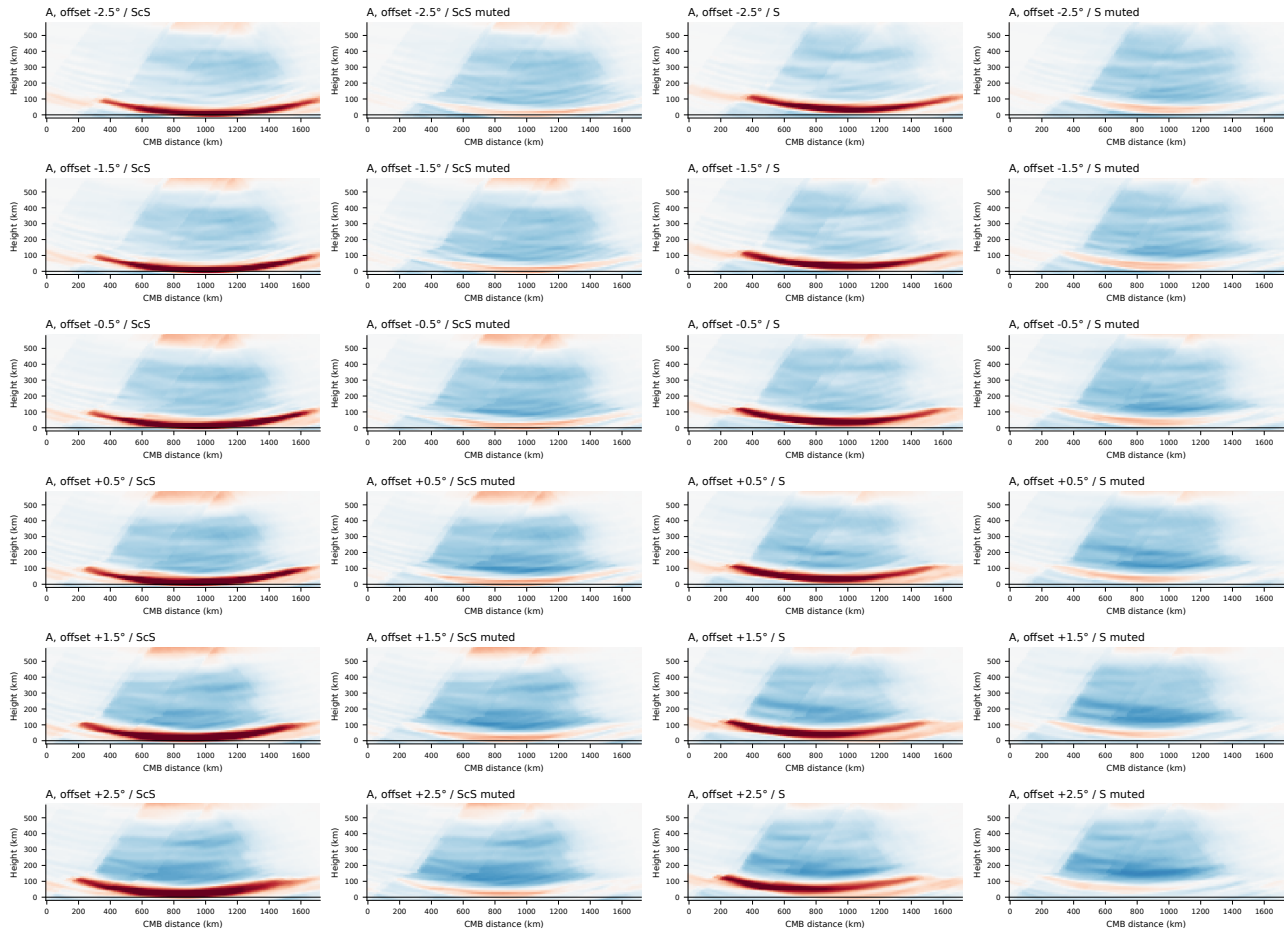


Figure 35: expanded; independent supplementary figure 3a counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

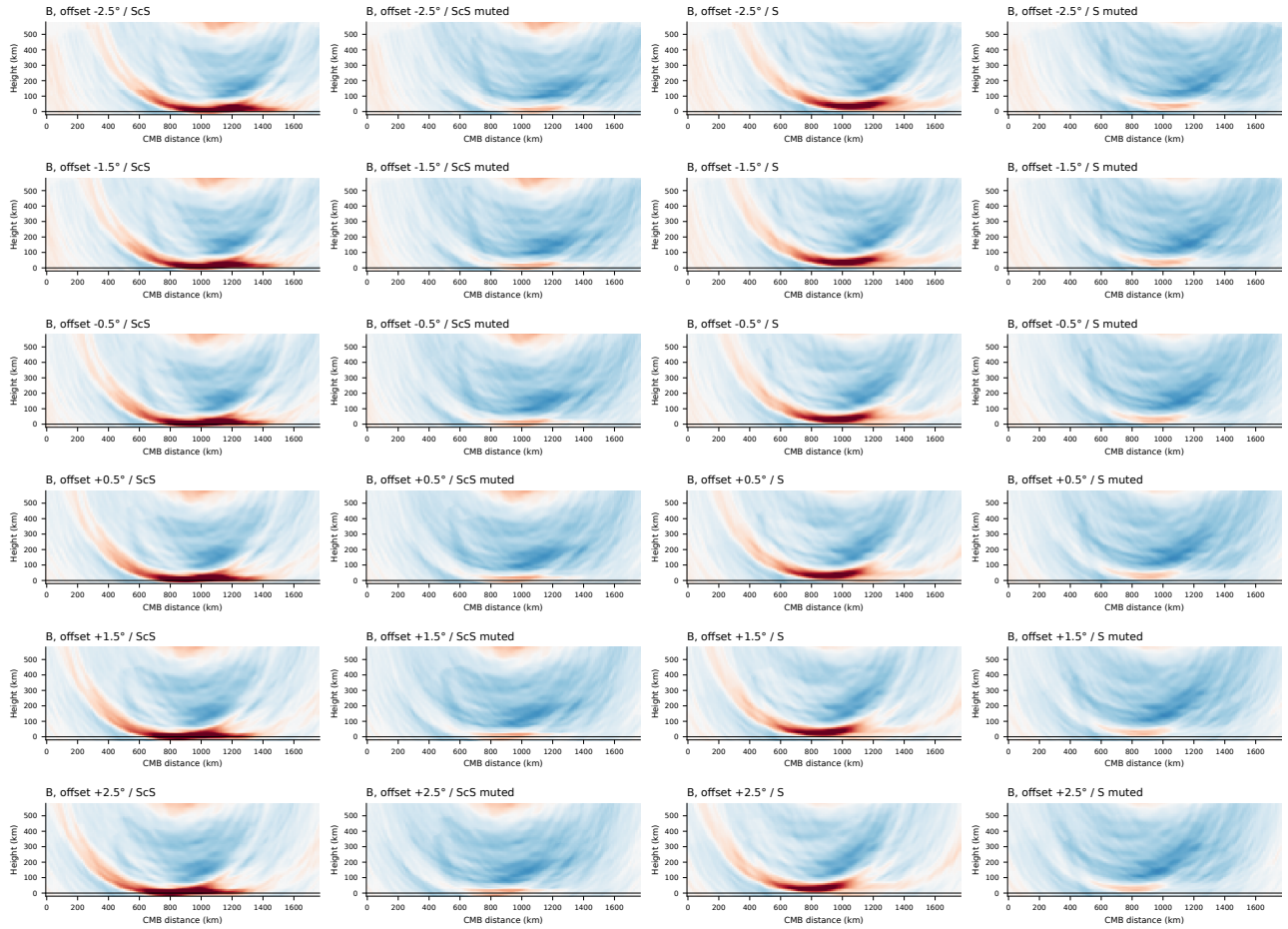


Figure 36: expanded; independent supplementary figure 3b counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

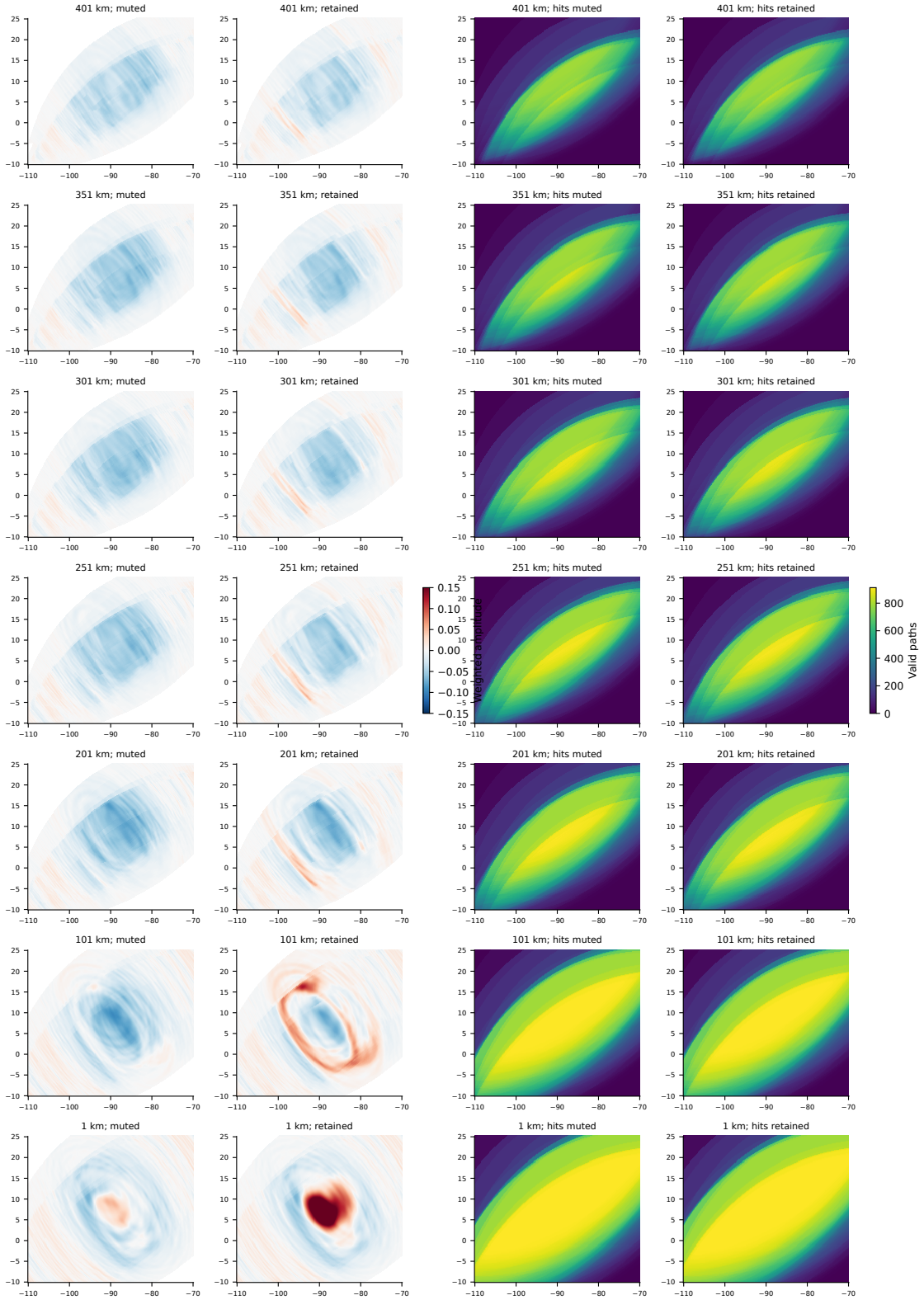


Figure 37: expanded; independent supplementary figure 5 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

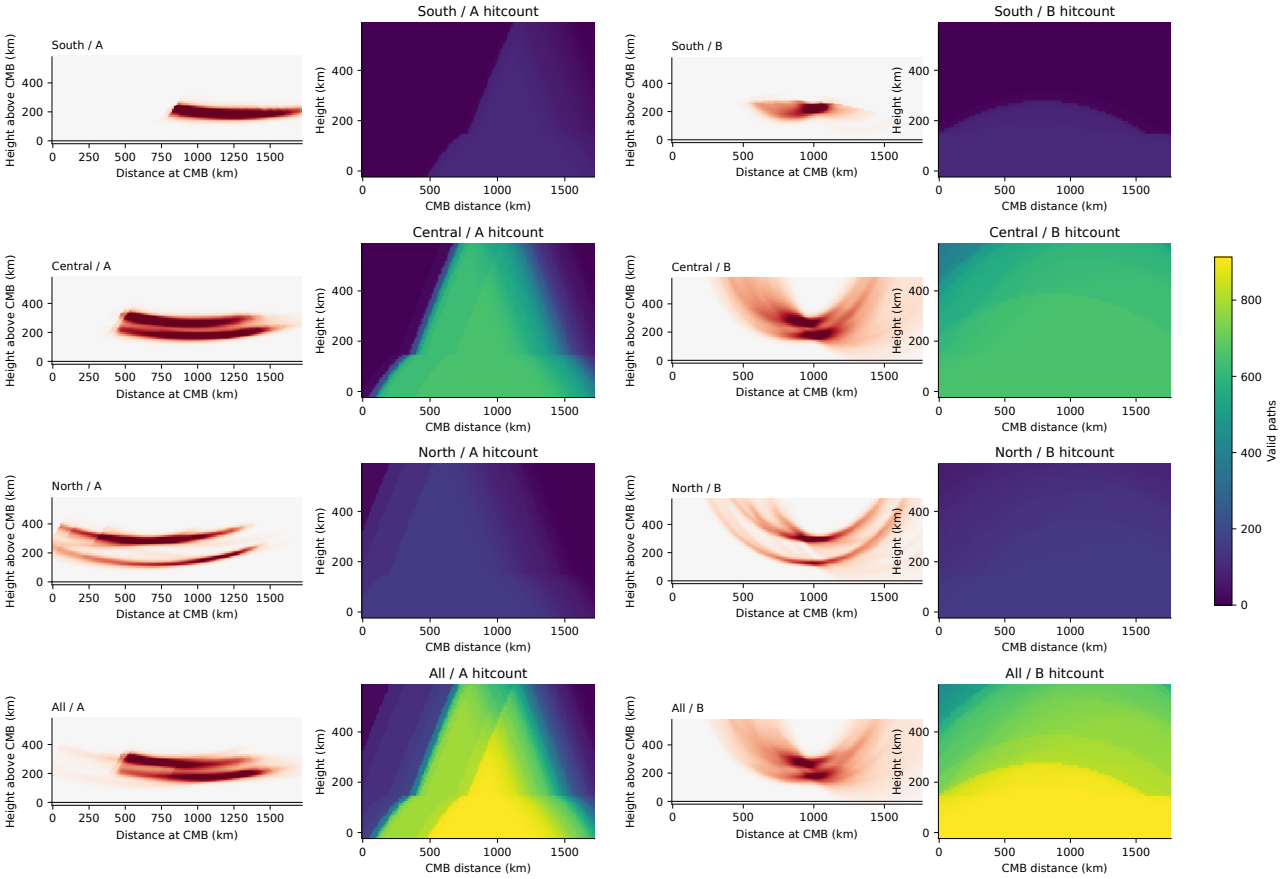


Figure 38: expanded; independent supplementary figure 6 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

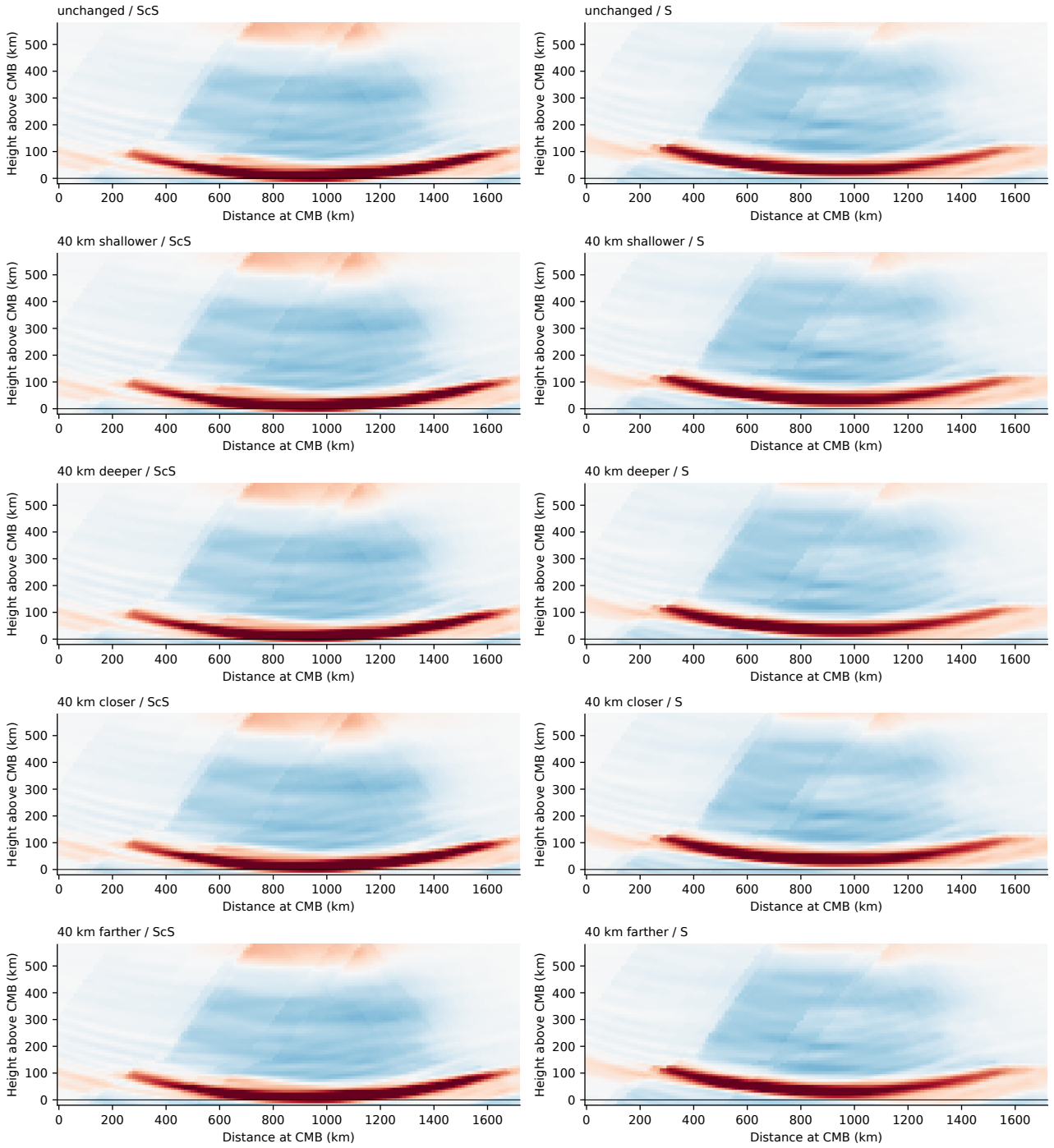


Figure 39: expanded; independent supplementary figure 10 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

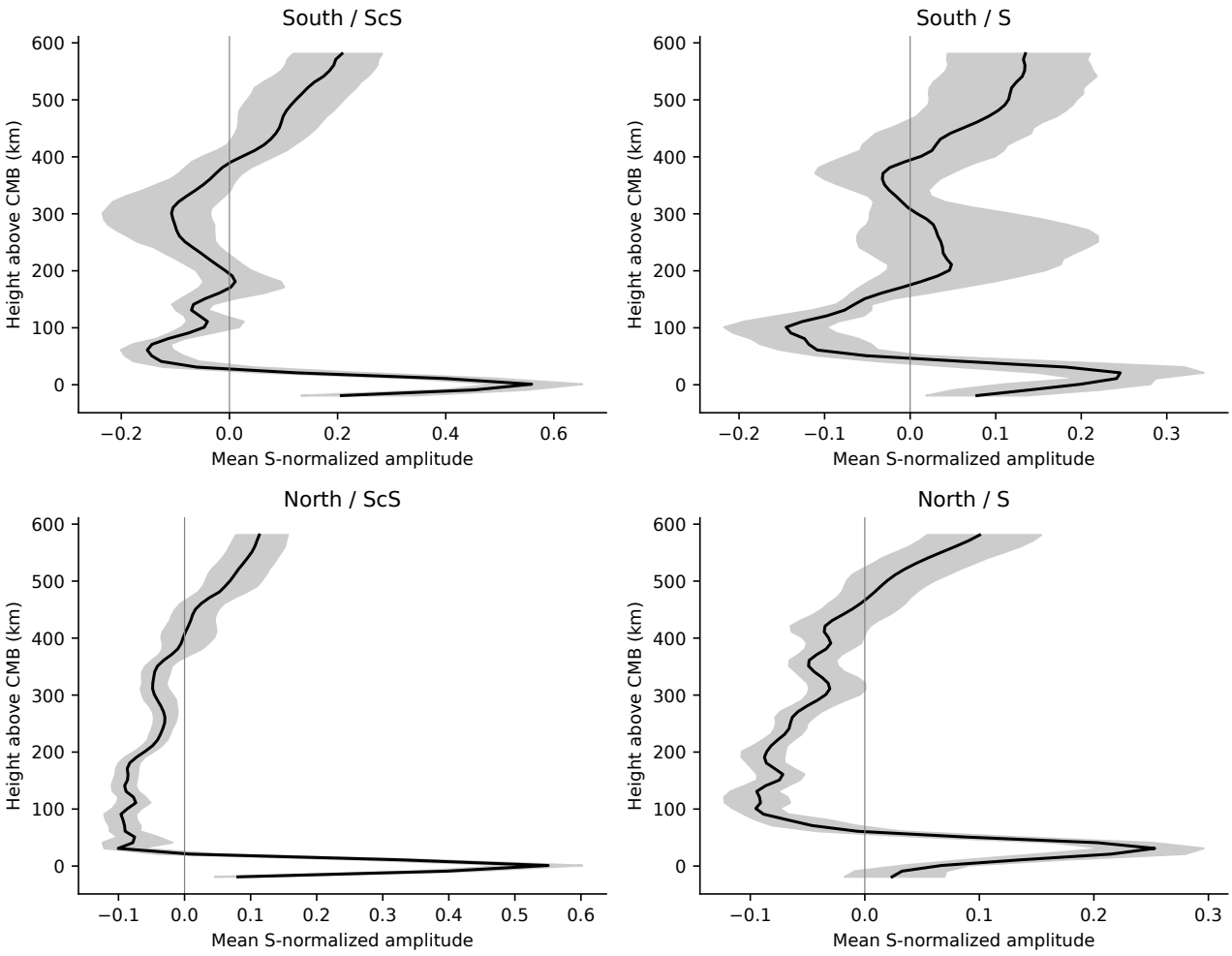


Figure 40: expanded; independent supplementary figure 11 counterpart. Definitions, assumptions, and panel status are given in the panel record. All numerical values come from saved arrays.

6 Panel-by-panel provenance and status

The complete machine-readable records are `report/panels.csv` and `report/panels.json`. Every row gives its quantity, documentation, parameters, data command, plot command, status, and limitation. The listing below includes every identified main and supplementary subpanel. The same status applies to each of the three generated dataset series unless a row explicitly names a separate schematic or unavailable calculation. Point-source resolution and source-location sensitivity are recomputed for each dataset geometry. In command strings, `DATASET` means `historical`, `modern_same`, or `expanded`. Parameters are fixed in `config.json` and described in the methods; no image-based tuning is used. "Partially reproduced" means independently generated but incomplete in inputs, method, or comparison, not a copied or digitized result. "Not reproduced" has no substituted numerical plot.

Command key (run from the project root):

A	`python scripts/acquire.py`
	`python scripts/process.py -workers 16`
M	`python scripts/migrate.py -dataset DATASET`
E	`python scripts/experiments.py -dataset DATASET`
V	`python scripts/validate.py -points -dataset DATASET`
L	`python scripts/sensitivity.py DATASET`
P	`python scripts/plot.py -dataset DATASET`

All E-only input-model and ray examples can be generated with `-synthetics-only`. Rows sharing a command still have separate entries in the accompanying complete CSV.

Panel	Quantity	Status	Data / plot commands
1 map	Event/station locations, ScS bounce points, profiles	partially reproduced	A, M / P
1 ray inset	S and ScS paths	partially reproduced	E / P
1 triplication inset	Sbc and Scd branch geometry	schematic redrawn	schematic / P
1 waveform inset	S, Scd+Sbc and ScS waveforms	partially reproduced	E / P
2a A	Signed migration amplitude, profile A; ScS retained	partially reproduced	A, M / P
2a B	Signed migration amplitude, profile B; ScS retained	partially reproduced	A, M / P
2b A	Signed migration amplitude, profile A; ScS censored	partially reproduced	A, M / P
2b B	Signed migration amplitude, profile B; ScS censored	partially reproduced	A, M / P
3 PREM	ScS-reference profile A in PREM	partially reproduced	A, M / P
3 UT	ScS-reference profile A in UT	not reproduced	unavailable / no substitute
3 UCB	ScS-reference profile A in UCB	not reproduced	unavailable / no substitute
3 CIT	ScS-reference profile A in CIT	not reproduced	unavailable / no substitute
4 upper	Folded slab and phase-boundary interpretation	schematic redrawn	schematic / P
4 lower	Profile A migration image	partially reproduced	A, M / P
S1 map	Geometry, mechanisms, Fresnel ellipses	partially reproduced	A, M / P
S2 ScS North	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 ScS Central	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 ScS South	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 ScS All	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 ScS hitcount	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 S North	Clustered profile or contributing ray count	partially reproduced	A, M / P

Panel	Quantity	Status	Data / plot commands
S2 S Central	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 S South	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 S All	Clustered profile or contributing ray count	partially reproduced	A, M / P
S2 S hitcount	Clustered profile or contributing ray count	partially reproduced	A, M / P
S3a A ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a A ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a A S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a A S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a B ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a B ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a B S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a B S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a C ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a C ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a C S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a C S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a D ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a D ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a D S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a D S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a E ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a E ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a E S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a E S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a F ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a F ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3a F S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3a F S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b U ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b U ScS censored	Parallel vertical migration section	partially reproduced	A, M / P

Panel	Quantity	Status	Data / plot commands
S3b U S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b U S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b V ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b V ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b V S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b V S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b W ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b W ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b W S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b W S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b X ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b X ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b X S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b X S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b Y ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b Y ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b Y S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b Y S censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b Z ScS retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b Z ScS censored	Parallel vertical migration section	partially reproduced	A, M / P
S3b Z S retained	Parallel vertical migration section	partially reproduced	A, M / P
S3b Z S censored	Parallel vertical migration section	partially reproduced	A, M / P
S4a A PREM	Velocity-model sensitivity: ScS retained	partially reproduced	A, M / P
S4a A UT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a A UCB	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a A CIT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a B PREM	Velocity-model sensitivity: ScS retained	partially reproduced	A, M / P
S4a B UT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a B UCB	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a B CIT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute

Panel	Quantity	Status	Data / plot commands
S4a C PREM	Velocity-model sensitivity: ScS retained	partially reproduced	A, M / P
S4a C UT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a C UCB	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a C CIT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a D PREM	Velocity-model sensitivity: ScS retained	partially reproduced	A, M / P
S4a D UT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a D UCB	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a D CIT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a E PREM	Velocity-model sensitivity: ScS retained	partially reproduced	A, M / P
S4a E UT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a E UCB	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a E CIT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a F PREM	Velocity-model sensitivity: ScS retained	partially reproduced	A, M / P
S4a F UT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a F UCB	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4a F CIT	Velocity-model sensitivity: ScS retained	not reproduced	unavailable / no substitute
S4b A PREM	Velocity-model sensitivity: ScS censored	partially reproduced	A, M / P
S4b A UT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b A UCB	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b A CIT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b B PREM	Velocity-model sensitivity: ScS censored	partially reproduced	A, M / P
S4b B UT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b B UCB	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b B CIT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b C PREM	Velocity-model sensitivity: ScS censored	partially reproduced	A, M / P
S4b C UT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b C UCB	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b C CIT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b D PREM	Velocity-model sensitivity: ScS censored	partially reproduced	A, M / P
S4b D UT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute

Panel	Quantity	Status	Data / plot commands
S4b D UCB	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b D CIT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b E PREM	Velocity-model sensitivity: ScS censored	partially reproduced	A, M / P
S4b E UT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b E UCB	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b E CIT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b F PREM	Velocity-model sensitivity: ScS censored	partially reproduced	A, M / P
S4b F UT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b F UCB	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4b F CIT	Velocity-model sensitivity: ScS censored	not reproduced	unavailable / no substitute
S4c A PREM	Velocity-model sensitivity: S reference, ScS retained	partially reproduced	A, M / P
S4c A UT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c A UCB	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c A CIT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c B PREM	Velocity-model sensitivity: S reference, ScS retained	partially reproduced	A, M / P
S4c B UT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c B UCB	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c B CIT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c C PREM	Velocity-model sensitivity: S reference, ScS retained	partially reproduced	A, M / P
S4c C UT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c C UCB	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c C CIT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c D PREM	Velocity-model sensitivity: S reference, ScS retained	partially reproduced	A, M / P
S4c D UT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c D UCB	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c D CIT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c E PREM	Velocity-model sensitivity: S reference, ScS retained	partially reproduced	A, M / P
S4c E UT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c E UCB	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c E CIT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute

Panel	Quantity	Status	Data / plot commands
S4c F PREM	Velocity-model sensitivity: S reference, ScS retained	partially reproduced	A, M / P
S4c F UT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c F UCB	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4c F CIT	Velocity-model sensitivity: S reference, ScS retained	not reproduced	unavailable / no substitute
S4d A PREM	Velocity-model sensitivity: S reference, ScS censored	partially reproduced	A, M / P
S4d A UT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d A UCB	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d A CIT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d B PREM	Velocity-model sensitivity: S reference, ScS censored	partially reproduced	A, M / P
S4d B UT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d B UCB	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d B CIT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d C PREM	Velocity-model sensitivity: S reference, ScS censored	partially reproduced	A, M / P
S4d C UT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d C UCB	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d C CIT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d D PREM	Velocity-model sensitivity: S reference, ScS censored	partially reproduced	A, M / P
S4d D UT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d D UCB	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d D CIT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d E PREM	Velocity-model sensitivity: S reference, ScS censored	partially reproduced	A, M / P
S4d E UT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d E UCB	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d E CIT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d F PREM	Velocity-model sensitivity: S reference, ScS censored	partially reproduced	A, M / P
S4d F UT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d F UCB	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S4d F CIT	Velocity-model sensitivity: S reference, ScS censored	not reproduced	unavailable / no substitute
S5a 400km	Horizontal migration amplitude	partially reproduced	A, M / P
S5a 350km	Horizontal migration amplitude	partially reproduced	A, M / P
S5a 300km	Horizontal migration amplitude	partially reproduced	A, M / P
S5a 250km	Horizontal migration amplitude	partially reproduced	A, M / P

Panel	Quantity	Status	Data / plot commands
S5a 200km	Horizontal migration amplitude	partially reproduced	A, M / P
S5a 100km	Horizontal migration amplitude	partially reproduced	A, M / P
S5a 0km	Horizontal migration amplitude	partially reproduced	A, M / P
S5b 400km	Horizontal migration amplitude	partially reproduced	A, M / P
S5b 350km	Horizontal migration amplitude	partially reproduced	A, M / P
S5b 300km	Horizontal migration amplitude	partially reproduced	A, M / P
S5b 250km	Horizontal migration amplitude	partially reproduced	A, M / P
S5b 200km	Horizontal migration amplitude	partially reproduced	A, M / P
S5b 100km	Horizontal migration amplitude	partially reproduced	A, M / P
S5b 0km	Horizontal migration amplitude	partially reproduced	A, M / P
S5c 400km	Valid ray hitcount	partially reproduced	A, M / P
S5c 350km	Valid ray hitcount	partially reproduced	A, M / P
S5c 300km	Valid ray hitcount	partially reproduced	A, M / P
S5c 250km	Valid ray hitcount	partially reproduced	A, M / P
S5c 200km	Valid ray hitcount	partially reproduced	A, M / P
S5c 100km	Valid ray hitcount	partially reproduced	A, M / P
S5c 0km	Valid ray hitcount	partially reproduced	A, M / P
S5d 400km	Valid ray hitcount	partially reproduced	A, M / P
S5d 350km	Valid ray hitcount	partially reproduced	A, M / P
S5d 300km	Valid ray hitcount	partially reproduced	A, M / P
S5d 250km	Valid ray hitcount	partially reproduced	A, M / P
S5d 200km	Valid ray hitcount	partially reproduced	A, M / P
S5d 100km	Valid ray hitcount	partially reproduced	A, M / P
S5d 0km	Valid ray hitcount	partially reproduced	A, M / P
S6a South amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6a South hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S6a Central amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6a Central hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S6a North amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6a North hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S6a All amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6a All hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S6b South amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6b South hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S6b Central amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6b Central hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S6b North amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6b North hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S6b All amplitude	Two-point synthetic migration resolution	partially reproduced	V / P
S6b All hitcount	Two-point synthetic migration resolution	partially reproduced	V / P
S7a	Observed deconvolved record section	partially reproduced	A / P
S7b	Layered-model synthetic waveforms	partially reproduced	E / P

Panel	Quantity	Status	Data / plot commands
S7c	Layered shear velocity profiles	partially reproduced	E / P
S8 model	Shear velocity step model	partially reproduced	E / P
S9a	Shear velocity step model	partially reproduced	E / P
S8 waveform 1	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S8 waveform 2	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S8 1D waveform	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9b North	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9b Central	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9b South	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9b All	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9c North	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9c Central	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9c South	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S9c All	Axisymmetric 2.5-D full-wave synthetics or their migration	not reproduced	unavailable / no substitute
S10 ScS unchanged	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 ScS shallower	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 ScS deeper	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 ScS closer	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 ScS farther	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 S unchanged	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 S shallower	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 S deeper	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 S closer	Profile A under 40 km source perturbations	partially reproduced	L / P
S10 S farther	Profile A under 40 km source perturbations	partially reproduced	L / P
S11 ScS North	One-dimensional double-array depth stack	partially reproduced	E / P
S11 ScS South	One-dimensional double-array depth stack	partially reproduced	E / P
S11 S North	One-dimensional double-array depth stack	partially reproduced	E / P
S11 S South	One-dimensional double-array depth stack	partially reproduced	E / P
S11 map	Double-array membership map	partially reproduced	A, M / P
S12 globe	UT2 tomography geometry and Vs cross-sections	not reproduced	unavailable / no substitute
S12 A	UT2 tomography geometry and Vs cross-sections	not reproduced	unavailable / no substitute
S12 B	UT2 tomography geometry and Vs cross-sections	not reproduced	unavailable / no substitute

Panel	Quantity	Status	Data / plot commands
S12 C	UT2 tomography geometry and Vs cross-sections	not reproduced	unavailable / no substitute

The ledger documentation is Hutko et al.'s main text and supplementary captions, with the processing papers cited above. Important family-specific differences are: independently selected parallel profile coordinates in S3/S4; assumed slice heights in S5; assumed point coordinates and profile endpoints in S6; incomplete station matching in S7a; no 2.5-D waveform solver in S8/S9; and no UT2 numerical tomography in S12. PREM S4 counterparts are displayed in S3 rather than duplicated under four different model headings. The full per-row explanatory notes are in the CSV.

7 Rebuilding, availability, and interpretation

From the project root, run `bash scripts/bootstrap.sh` once, then `bash rebuild.sh -online`. With the downloaded archive cache present, `bash rebuild.sh` runs without acquisition. Both procedures regenerate numerical figures and compile this report plus one event supplement per dataset. The build exits on a failed validation or LaTeX command. Execution logs and resource measurements are in `logs/`. The `README.md` documents fresh-output removal, file formats, exclusions, and publication commands.

The calculations do not independently establish the published step or the folded slab interpretation. The retained signal population and omitted receiver corrections are sufficient reasons to expect differences, but do not prove which discrepancy dominates. A stronger reconstruction requires confirming the inferred historical event membership, recovering station-specific splitting values, and verifying phase and wavelet selections, followed by full-wave tests. Expanded sampling adds useful observations but does not by itself remove these systematic uncertainties.

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